# Particles and Nuclei 2

- 4/9/2026
- Announcements:

- HW 5 finalized due date Tues

- Office Hours Tomorrow and (probly) Monday 11am

- Reading: see email from Sat 4/4

# More about the Color Factor in Prob 1 b

- Not much new to add today about this but just some further tips if you DO want to go for the full calculation for qg→qg color factor

- To review, the main part of that problem is meant to be part 1, where you should look to the Halzen and Martin (HMJ) result as the thing your shooting for (u/s+s/u form) NOT the form given in Griff 7.47.
- **For that part alone** there are a couple more hints: one new one, but first repeating yesterday's msg in the canvas announcement:
  - You SHOULD use the result in Halzen and Martin (in the slides) that the cross terms of the matrix element (e.g. M1M2*) are 0, but you should try to prove it, if you can't show your work and still just use the result . This RESULT IS ZERO MOSTLY/ONLY BECAUSE YOU SHOULD BE SETTING _ALL_ THE MASSES TO ZERO.
- The new comment from yesterday:
  - For showing the cross terms e.g. M1M2* are 0 as stated in HM, you may need the following relation for Mendalstam vars from exercises in Ch. 3 $s+t+u= m^2 \approx 0$ (for us)

# Next slide:

- For reference the next slide again contains the main instructions regarding part 1b I already posted

# ("Help") For gluon Compton color factor

- Part 1b) I will accept any answer **with ANY kind of physics-based logic/justification** for about 95% of the credit [for the color factor part/cross section part—all but ~0.5-1pt].
- For the remaining (~1 pt) credit, I will apply only a little subjectivity into the effort put into your justification (which doesn't have to be great or correct but should involve some reference to the general method of calculating CF's (#q/g, averaging etc.)
- I will discuss a few more details next time but for example
  - If you choose to do the easier color factor for q+photon→q+g or the reverse, make sure you choose a quark flavor and then also include the extra factor for the fractional charge of the quark (or use an average over all flavors—any averaging w/ reasonable justif is OK)
  - In that case I recommend for the color using the result in Halzen&Martin and explaining it based on the color counting method described there

  - For the gq→gq one option is to also use that color counting method [and don't worry if correct] whether this yields the correct answer or not
  - You can also just try to follow what Griffiths provides, I will not be very critical—this is what I give more info on in next slides

# Read this!

- The method I'm referring to is described here:

(THIS IS IN HANDOUT p.40 of pdf, 211 of book)

$$\overline{|\mathfrak{M}|^2} = 32\pi^2\left(e_i^2\alpha\alpha_s\right)\frac{4}{3}\left(-\frac{\hat{t}}{\hat{s}} - \frac{\hat{s}}{\hat{t}} + \frac{2\hat{u}Q^2}{\hat{s}\hat{t}}\right). \tag{10.17}$$

The invariant variables are denoted $\hat{s}, \hat{t}, \hat{u}$ to indicate that we are considering a parton subprocess.

The factor $\frac{4}{3}$ takes into account the summation/averaging over final/initial colors. It can readily be deduced from counting the color lines shown in Fig. 10.6. Each color line can take one of three possible colors, so there are $(3 \times 3 - 1)$ different configurations, if we subtract the color singlet configuration as explained in Section 2.15. We now average over the three initial quark colors and obtain $\frac{8}{3}$. However, this result has to be divided by a factor 2 because of the unfortunate historical definition of $\alpha_s$ (see Section 2.15).

Most simple QCD diagrams are exactly analogous to QED diagrams, and the QCD cross section is obtained by the replacement

$$\alpha^n \to C_F\alpha_s^n, \tag{10.18}$$

where $n$ is the number of quark–gluon (or gluon–gluon) vertices in the diagram. The color factor $C_F$ is obtained by simply summing and averaging in much the same way as we do for spin. However, without changing the cross section, the color factor can be altered by a redefinition of $\alpha_s$. For example,

$$\alpha_s^n C_F \to (2\alpha_s)^n (C_F/2^n) = (\alpha_s')^n C_F'. \tag{10.19}$$

The conventional definition of $\alpha_s$ corresponds to the second convention in (10.19). The final rule to obtain the conventional color factor is simple: "count color on your fingers" and divide the result by $2^n$. This is why we divided the color factor $\frac{8}{3}$ of Fig. 10.6 by $2^1$.

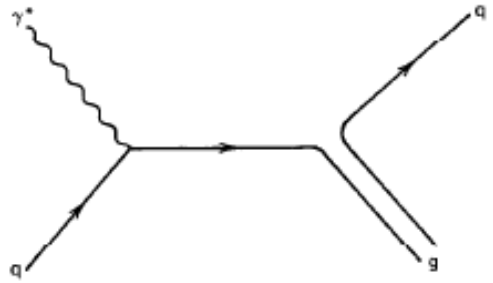



**Fig. 10.6** Color lines for $\gamma^* q \to qg$.

$$\overline{|\mathcal{M}|^2} = 32\pi^2\left(e_i^2\alpha\alpha_s\right)\frac{4}{3}\left(-\frac{\hat{t}}{\hat{s}} - \frac{\hat{s}}{\hat{t}} + \frac{2\hat{u}Q^2}{\hat{s}\hat{t}}\right). \qquad (10.17)$$

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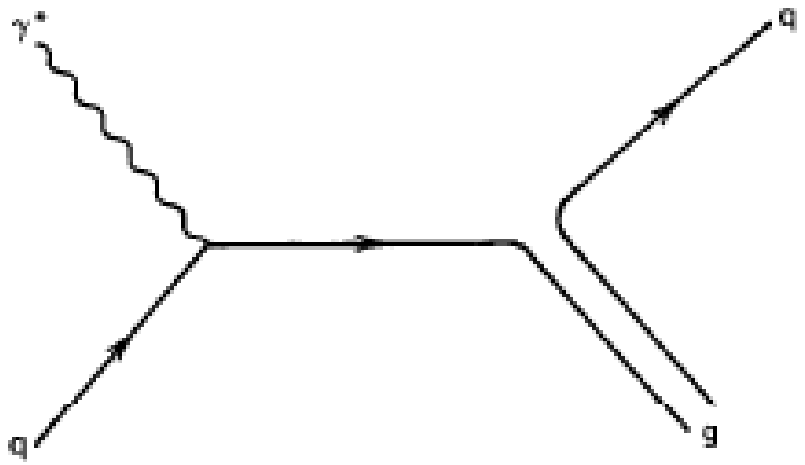



**Fig. 10.6** Color lines for $\gamma^* q \to qg$.

your fingers" and divide the result by $2^n$. This is why we divided the color factor $\frac{8}{3}$ of Fig. 10.6 by $2^1$.

# H&M Various Compton Processes

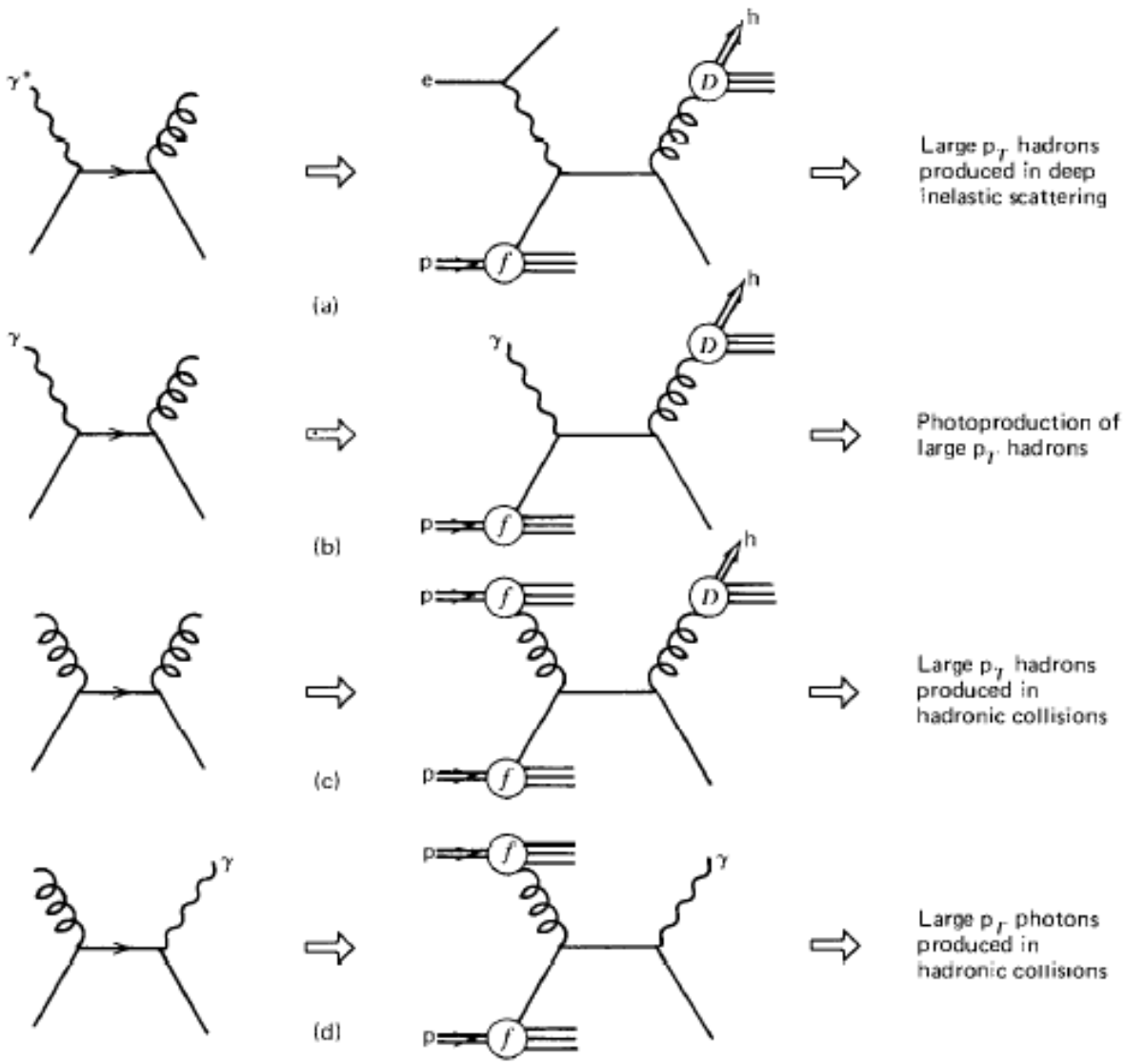



**Fig. 11.13** Various observable processes which contain γ (or gluon)–quark Compton scattering, or the crossed reaction, as a parton subprocess; *f* and *D* denote structure and

- help for color factors for gluon here to be discussed later

$$\overline{|\mathcal{M}|^2} = 32\pi^2\left(e_i^2\alpha\alpha_s\right)\frac{4}{3}\left(-\frac{\hat{t}}{\hat{s}} - \frac{\hat{s}}{\hat{t}} + \frac{2\hat{u}Q^2}{\hat{s}\hat{t}}\right). \tag{10.17}$$

The invariant variables are denoted $\hat{s}, \hat{t}, \hat{u}$ to indicate that we are considering a parton subprocess.

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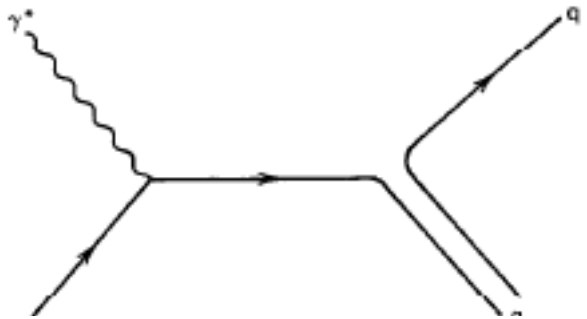
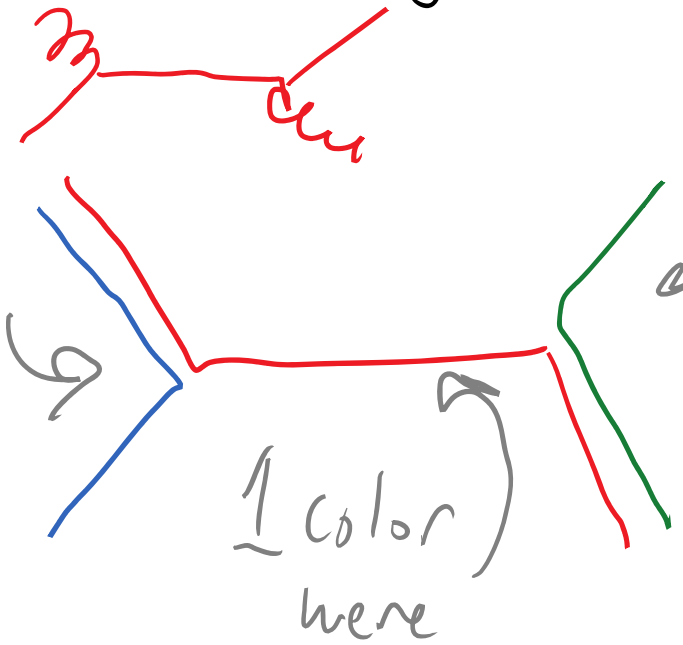


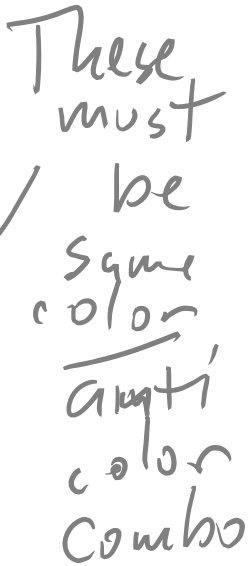

Fig. 10.6. Color lines for $\gamma^* q \to qg$

# New Sections Help on Color Factor for Part 1b

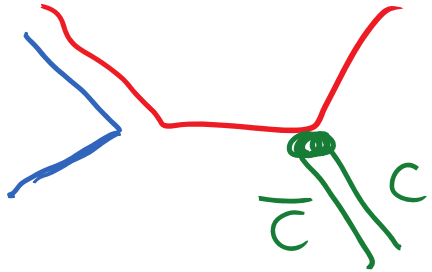
- One thing to borrow from color reading on previous slides, no matter how you approach the 1b color factor

- the color "flow" logic :which constrains the number of possible incoming and outgoing colors since color is conserved and gluons contain a color and anti-




Sidenote:



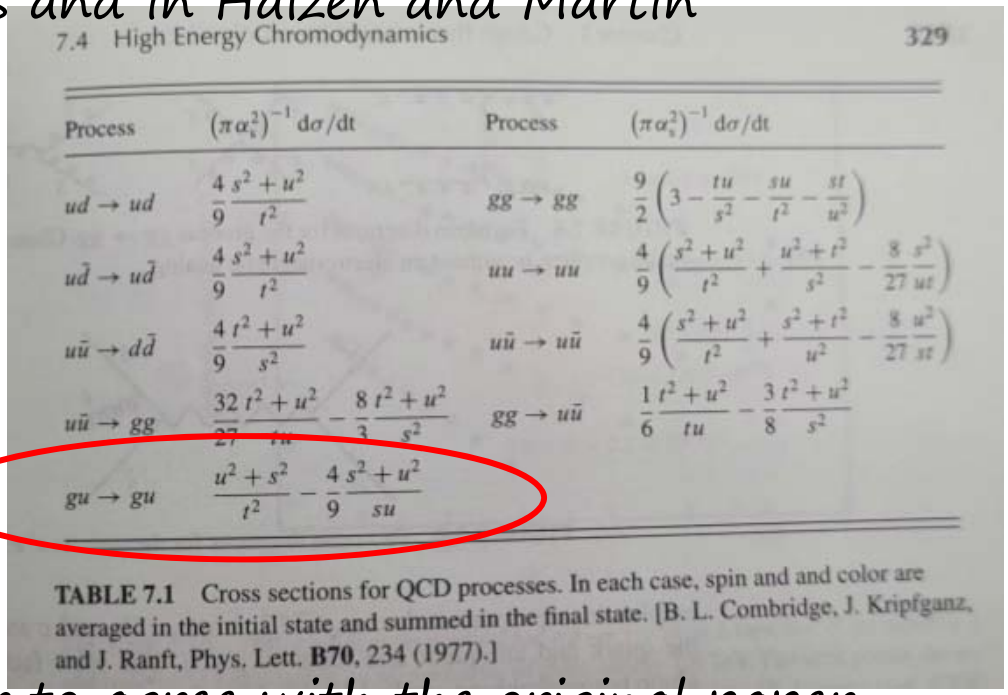

Another possiblity?
you may ignore this case regardless

So there are definitely less than 3 x3 x3 possible incoming and outgoing color possibilities and not just less by minus 1 accounting for 3^2-1 for exclusion of gluon octet

# 1b gq→gq: the answer

- For "going for it" and trying for the full factor for gluons on both ends of the diagrams, rather than replacing one with the photon vertex.
- First the real answer? The most common value given in the literature is 4/9*for the factor squared implying the factor is 2/3, or maybe 1/3 if they were using a convention for $\alpha_S$ without the factor of 2 discussed in class and in Halzen and Martin

- (From textbook *Particle Physics* Duncan Carlsmith):



7.4 High Energy Chromodynamics 329

| Process | $(\pi\alpha_s^2)^{-1}\,d\sigma/dt$ | Process | $(\pi\alpha_s^2)^{-1}\,d\sigma/dt$ |
|---|---|---|---|
| $ud \to ud$ | $\frac{4}{9}\frac{s^2+u^2}{t^2}$ | $gg \to gg$ | $\frac{9}{2}\left(3-\frac{tu}{s^2}-\frac{su}{t^2}-\frac{st}{u^2}\right)$ |
| $u\bar{d} \to u\bar{d}$ | $\frac{4}{9}\frac{s^2+u^2}{t^2}$ | $uu \to uu$ | $\frac{4}{9}\left(\frac{s^2+u^2}{t^2}+\frac{u^2+t^2}{s^2}-\frac{8}{27}\frac{s^2}{ut}\right)$ |
| $u\bar{u} \to d\bar{d}$ | $\frac{4}{9}\frac{t^2+u^2}{s^2}$ | $u\bar{u} \to u\bar{u}$ | $\frac{4}{9}\left(\frac{s^2+u^2}{t^2}+\frac{s^2+t^2}{u^2}-\frac{8}{27}\frac{u^2}{st}\right)$ |
| $u\bar{u} \to gg$ | $\frac{32}{27}\frac{t^2+u^2}{tu}-\frac{8}{3}\frac{t^2+u^2}{s^2}$ | $gg \to u\bar{u}$ | $\frac{1}{6}\frac{t^2+u^2}{tu}-\frac{3}{8}\frac{t^2+u^2}{s^2}$ |
| $gu \to gu$ | $\frac{u^2+s^2}{t^2}-\frac{4}{9}\frac{s^2+u^2}{su}$ | | |

**TABLE 7.1** Cross sections for QCD processes. In each case, spin and and color are averaged in the initial state and summed in the final state. [B. L. Combridge, J. Kripfganz, and J. Ranft, Phys. Lett. **B70**, 234 (1977).]

*Note in this ref, the 4/9 includes all factors from the cross section (like 1/16 or 1/64) AS WELL as any factor from the QED M^2. Other references get 4/9 there but not incl the same other factors

- However this doesn't appear to agree with the original paper reference given underneath the result→ and I cannot get that result myself trying it different ways.
- By all means use this number if you like, but I would still like some justified attempts however little to go with that.

# 1b gg→gg: Following Feynman rules

- Following Feynman rules same as QED Compton but including color objects:

$$\mathcal{M}_{s\ \text{Channel}} = c_4^{\dagger}\bar{u}_4\gamma^{\mu}\lambda^{\alpha}\varepsilon_3 a_3^{\alpha}\frac{(g_s/2)^2}{q^2-m}\lambda^{\beta}\varepsilon_2 a_2^{\beta}\gamma^{\nu}\lambda^{\beta}u_1 c_1$$

$$= \frac{g_s^2/4}{s}\, a_3^{\alpha} a_4^{\beta}\, c_4^{\dagger}\lambda^{\alpha}\lambda^{\beta}c_1 * [\text{QED result}]$$

color factor part

- At this point we need to sum over initial and final state particles and (average on initial colors sum, but also divided by "color counting" factors (e.g. 3 for quarks, 8 for gluons but see last slide)
- First about the gluon **a** vectors/ factors. Most references don't have them (ie don't even include them in the Feynman rules. ) I recommend doing that too. ie ignore. As given by Griffiths but not fully defined, they seem to me to only provide to require that e.g. in this case $\delta_{ab}$ (they are unit vectors dotted with each other) which says that incoming and outgoing gluon must be the same color state which couldn't be completely correct.

# 1b gg→gg: Following Feynman rules

- Following Feynman rules same as QED Compton but including color objects:

$$\mathcal{M}_{s\,\text{Channel}} = c_4^{\dagger}\bar{u}_4\gamma^{\mu}\lambda^{\alpha}\varepsilon_3 a_3^{\alpha}\frac{i g_s/2}{q^2-m}\, a_2^{\beta}\varepsilon_2\gamma^{\nu}\lambda^{\beta}u_1 c_1$$

$$= \frac{g_s^2/4}{s}\, a_3^{\alpha} a_4^{\beta}\, c_4^{\dagger}\lambda^{\alpha}\lambda^{\beta}c_1 * [\text{QED result}]$$

color factor part

- A not too crazy option is to implement the matrix multiplications and expand the color vectors as sum of (1,0,0)*X + (010) * X … and use a matrix package like gsl in c++ or mathematica or python to do a bunch of the sums.

- For the sums we in these cases just choose to sum over all 8 gluon states incoming and outgoing just to simplify things, this also leads to some algebraic simplifications on the next slide

# 1b gg→gg: Following Feynman rules

- Following Feynman rules same as QED Compton but including color objects:

$$\mathcal{M}_{s\text{ channel}} = c_4^\dagger \bar{u}_4 \gamma^\mu \lambda^\alpha \varepsilon_3 \, q_3^\alpha \, \frac{i g_s/\sqrt{2}}{q^2 - m} \, q_2^\beta \varepsilon_2 \gamma^\nu \lambda^\beta u_1 c_1$$

$$= \frac{g_s^2/4}{s} \, q_3^\alpha q_4^\beta \, c_4^\dagger \lambda^\alpha \lambda^\beta c_1 * [\text{QED result}]$$

color factor part

- Simplify notation : $c_4^\dagger \lambda \lambda c_1 \Rightarrow (\lambda^\alpha \lambda^\beta)_{ij}$
- indexes i j indicate which colors the c's were. When squaring for M^2 the colors (incl gluon) must match so we get

$$|\mathcal{M}_s|^2 \propto (\lambda^\alpha \lambda^\beta)_{ij} (\lambda^\beta \lambda^\alpha)_{ji}$$

(Summing) $\Rightarrow$

$$= \mathrm{Tr}(\lambda^\alpha \lambda^\beta \lambda^\beta \lambda^\alpha)$$

# Traces for Colors

- There are many trace theorems come up including these ones.

$$\mathrm{Tr}\left(\lambda^{\alpha}\lambda^{\beta}\right) = 2\delta_{\alpha\beta}$$

$$\mathrm{Tr}\left(\lambda^{\alpha}\lambda^{\beta}\lambda^{\beta}\lambda^{\alpha}\right) = 16/3$$

$$\mathrm{Tr}\left(\lambda^{\alpha}\lambda^{\beta}\lambda^{\alpha}\lambda^{\beta}\right) = 32/3$$

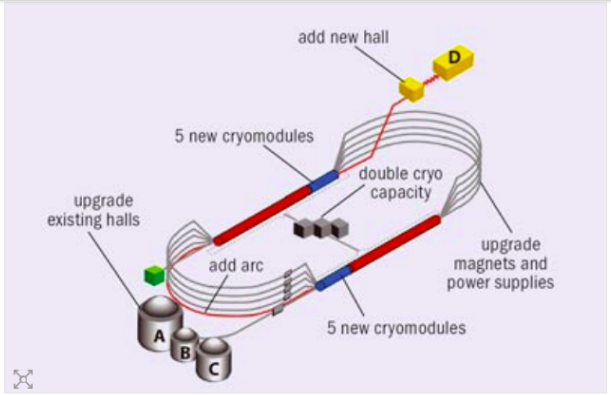
- Another helpful identity here:

$$\sum_{\alpha} \lambda^{\alpha}\lambda^{\alpha} = \frac{16}{3}\,\mathbb{1}_{3\times 3}$$

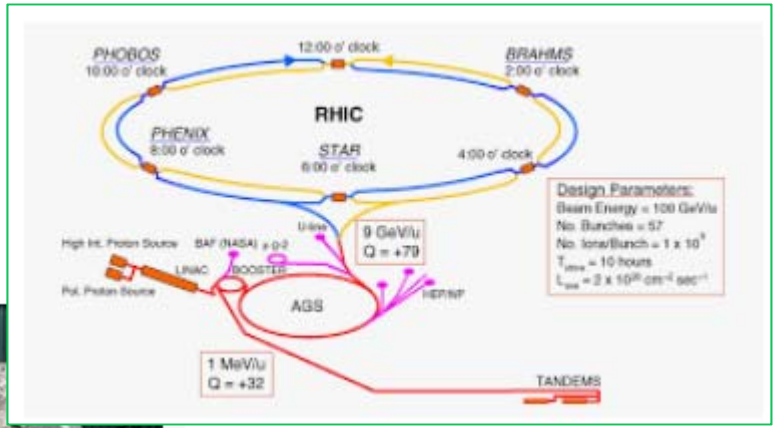
# Big Synchotrons → Colliders

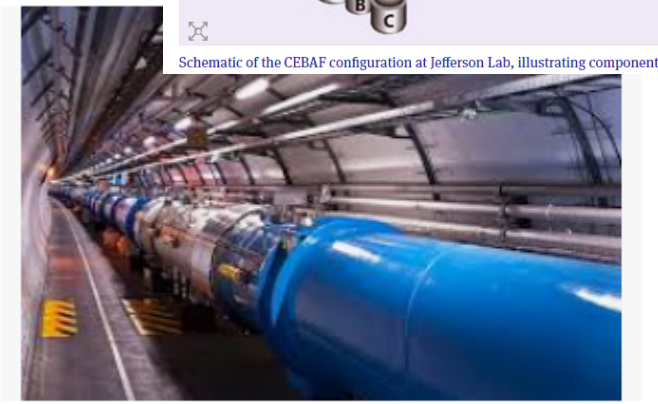
- "Simplest : Cyclotrons truly circular
  - V (in region) B continually adjusted
  - Most common at many universities < ~1965
- More common/More important
  - Combinations of "LINAC"s and Magnet Trains

sciencedirect.cor

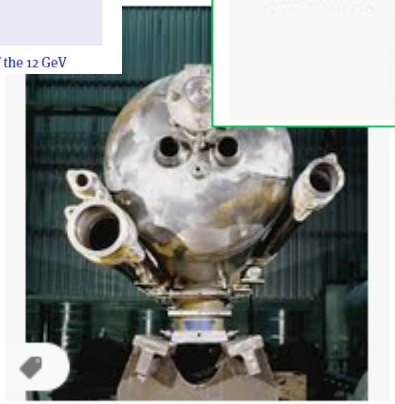



Schematic of the CEBAF configuration at Jefferson Lab, illustrating components of the 12 GeV



Largest Particle Accelerator ...
scientificamerican.com



Lhc Particle Accelerator Ma...
pixels.com · In stock

# Flux, Luminosity Bunches

- The **focusing part (tranverse size)** determines the other most important part of an accelerator Flux/ LUMINOSITY
- **Smaller Area higher Flux/Lumi**

**Flux**: A beam of incoming pointlike particles has flux density $j$ (#/m$^2$/sec) uniformly spread over an area $A_b$ [m$^2$] with total flux $J = jA$, [#/sec]. The flux density can be written $j = n_L v_L$, where

$$L = \frac{1}{\sigma}\frac{dN}{dt}.$$

- For colliders **Luminosity** ~same units but like effective flux taking into account both beam fluxs – x.s. in def unavoidable
- Bunching (part of focusing) big driver

$$\mathcal{R} = \sigma\mathcal{L} \quad \text{with } \mathcal{L} = \mathrm{f}n\frac{N_1 N_2}{A}$$

where the quantities associated with the luminosity $\mathcal{L}$ are

f = frequency of revolution of bunches
$n$ = number of bunches per beam
$N_1, N_2$ = number of particles per bunch in beams 1,2
$A$ = area of beam
$\sigma$ = cross-section for interaction
(computed from underlying the

Usually given in "inverse" X barns

E.g. perhaps 1 fb$^{-1}$ of data taken in one year

$\mathcal{L}_{1\,\mathrm{bcw}} \propto \frac{N}{A \cdot s}$

- Integrated Luminosity

$$L_{\mathrm{int}} = \int L\,dt.$$

# Luminosity: Stochastic/Electron Cooling

- Lumi formula → Inversely prop to beam area A

$$\mathcal{R} = \sigma \mathcal{L} \quad \text{with } \mathcal{L} = \mathrm{f} n \frac{N_1 N_2}{A}$$

- We discussed betatron oscillations/defocusing as one source of transverse beam
- Another is "heating" : thermal drift outward

- An important ~recent advance to avoid that is **Stochastic cooling** → described in Mann
- Important upgrade to RHIC
- Cooling also planned for EIC → **"Electron Cooling" like laser cooling → another beam of electrons around beam!**

# Lumi Fixed Target

- For fixed target like at Jlab, obviously dense (regular) matter used as target

- Lumi formula often related directly to target density and thickness

$$\mathcal{L} = J n_t \Delta x = n_b v_b A_b n_t \Delta x$$

- B.c. densities (nt) are so much higher than beam densities Lumi obviously higher at Fixed Target

- INTENSITY FRONTIER VS Energy Frontier

- Jlab Lumi's 10^37-39 RHIC : 10^32

- LHC HL → 10^35

# Scintillators

- MOST IMPORTANT DETECTOR TYPE IN Medium and High energy particle physics.

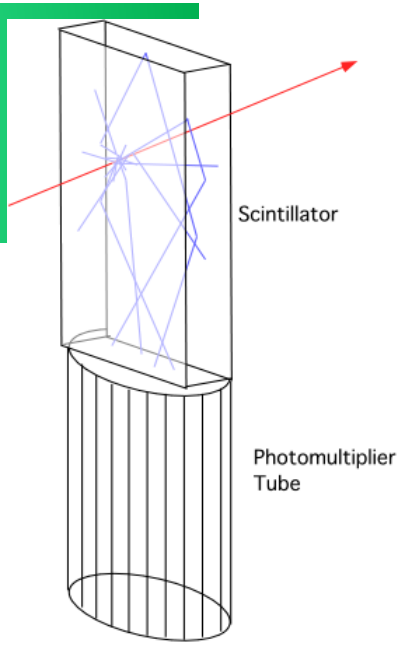
- **ALL PURPOSE TRIGGER!!!!**

- Translucent Material that emits visible or near visible light when charge particle passes through.
- Light is collected by light detection devices converted into voltage/current signal

- Mostly for triggering just yes/no.
  - (at low energy can give precise energy measurements)

- At high energy these can also provide the ~best timing resolution of all detectors, so also used for that (mainly for PID)
  - time → velocity → p → combined with E meas, PID.

# Scintillators



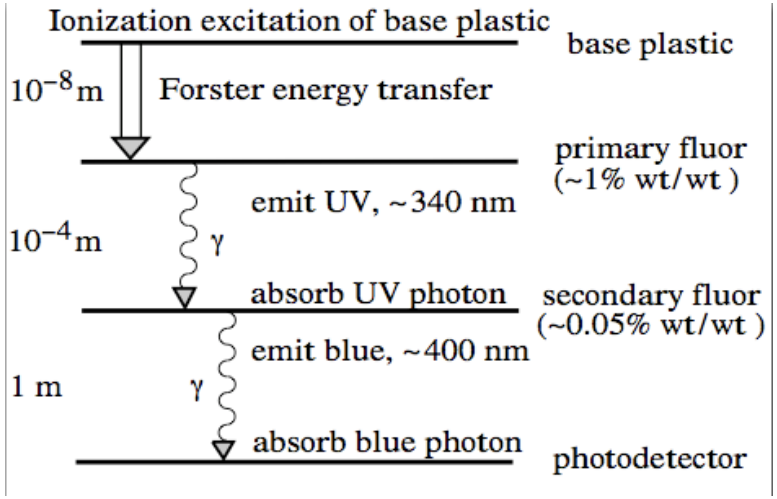

**Scintillation Counters** are probably the most widely used detectors in Nuclear and High Energy Physics.

Scintillator material are special material that

-emit light when traversed by energetic particles and

-can shift the wavelength of this light

to be harnessed by PMTs



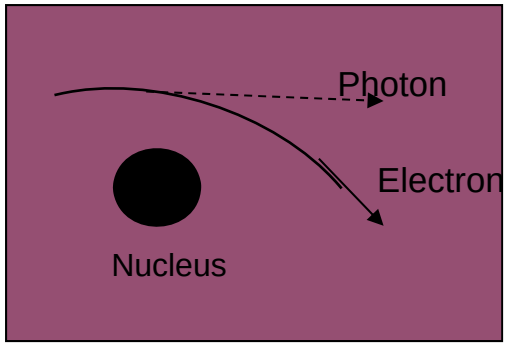

They can be solid, liquid (even gas)

They can be molded in all kind of shapes

# Charged Particles: Bremsstrahlung

**In principle include (not of HW) but in reality not affecting for practical cases – except e showering**

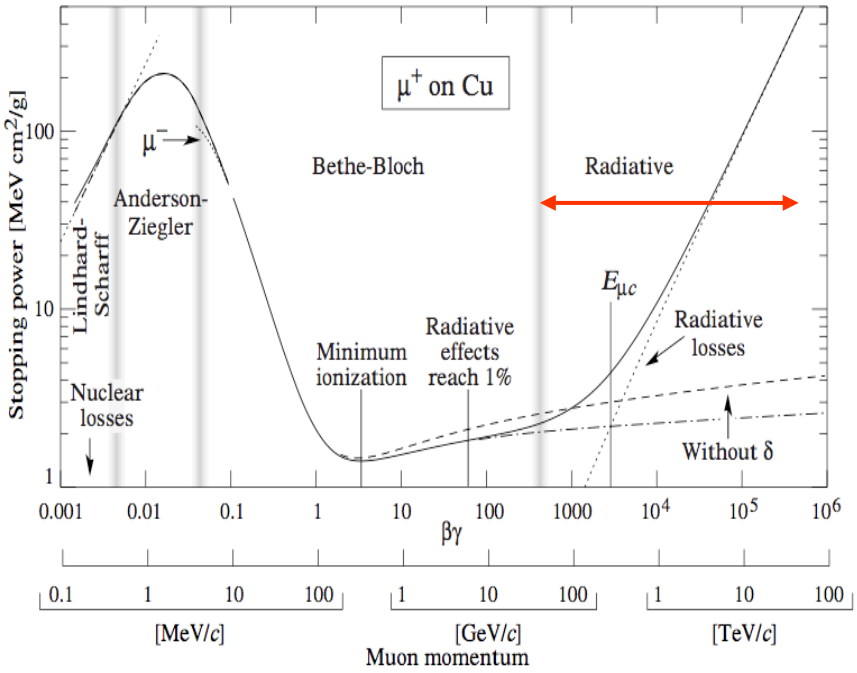
- Radiation of real photons in the Coulomb field of a nuclei of the absorber.




$$-\frac{dE}{dX} = 4\alpha \frac{\rho N_A}{A} Z(Z+1) r_e^2 \ln(183 Z^{-1/3}) E$$

- Define $X_0$ the radiation length: length during which the particle looses a fraction *e* of its initial energy

$$-\frac{dE}{E} = \frac{dX}{X_0}$$



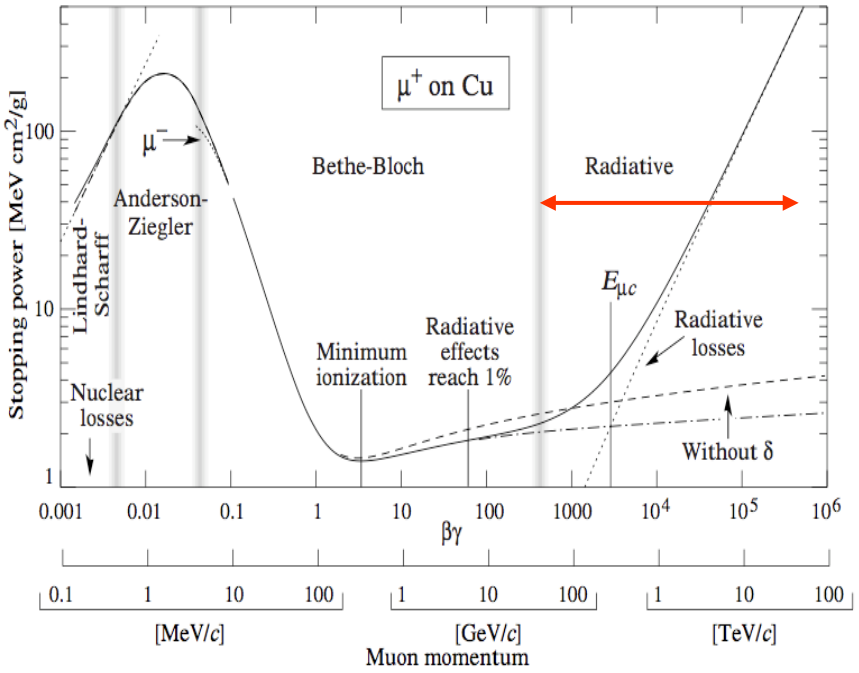

- Critical energy ($E_{\mu c}$) is the energy at which Bremstrahlung loss equal ionization loss

  $E_{\mu c}$(e- Cu)=20 MeV

  $E_{\mu c}$(μ- Cu)=800 GeV

# Showering → Radiative Loss

- Mostly we are just dealing w/ electrons even for photons
- Radiation length in EM cal
- ~steel/W (tungsten) ~2 cm




- Electrons Radiate strongly> MeV
- Resulting γ : pair production
- ...Showering
- Define $X_0$ the radiation length: length during which the particle looses a fraction e of its initial energy

$$-\frac{dE}{E} = \frac{dX}{X_0}$$

Estimate energy "leakage" (non released energy) if detector thickness is 30 cm

LEADS TO EXPONENTIAL prob of survival for w/ depth x (γ) or energy deposition

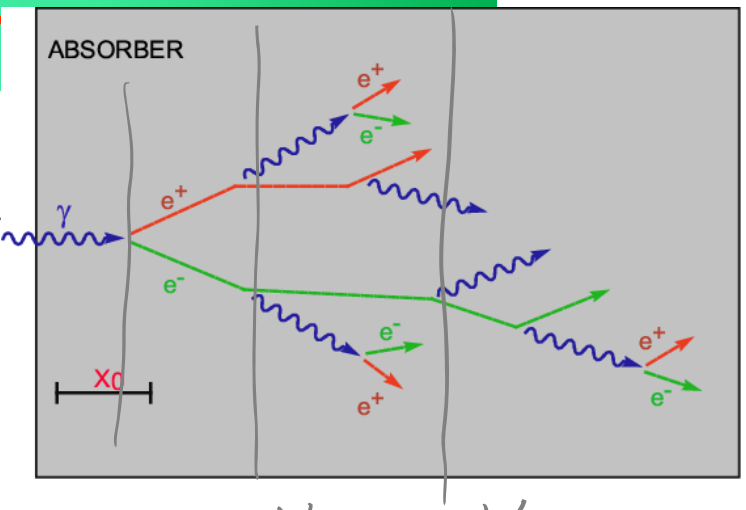
$$E / N \propto e^{-X/X_0}$$

# Particle Detectors: Calorimeter

*Used to measure energy.*

*"Long" matter: EM alternating Brems. &*

*Pair production → "Shower"*




*Suppose an initial photon of energy $E_0$*

*After t radiation lengths, the average energy of secondary is:* $E(t) = E_0 / 2^t$

*The shower stops when* $E(t_{\max}) = E_c$ $\qquad t_{\max} = \dfrac{\ln(E_0/E_c)}{\ln 2} \sim 10X_0$

*The number of secondaries is* $N_{\max} = \exp(t_{\max} \ln 2) = E_0 / E_c$

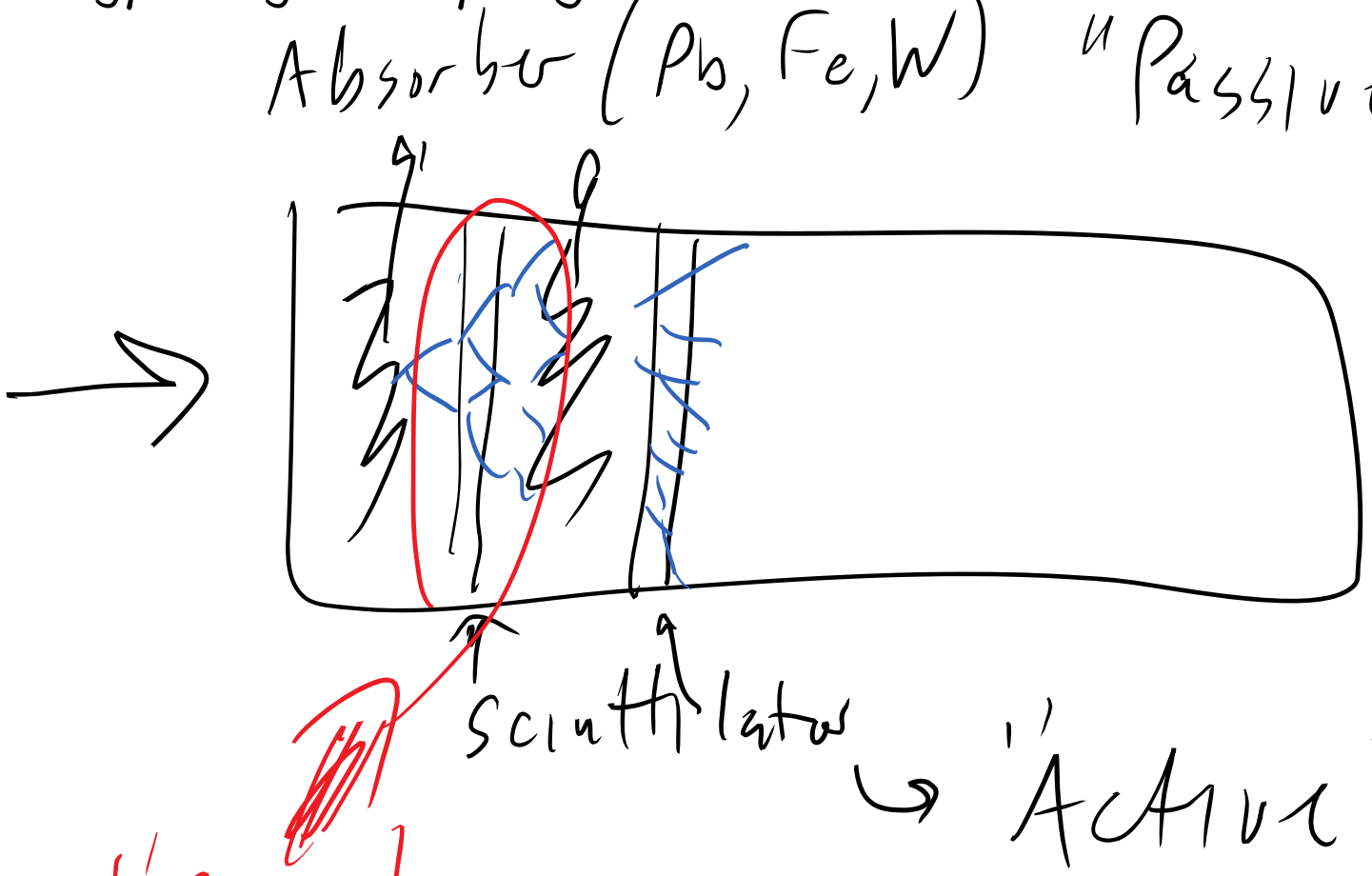
*The energy resolution of the calorimeter therefore goes as* $\dfrac{\Delta E_0}{E_0} = \dfrac{1}{\sqrt{E_0}}$

**Energy RESOLUTION Dependent on number of shower particles**

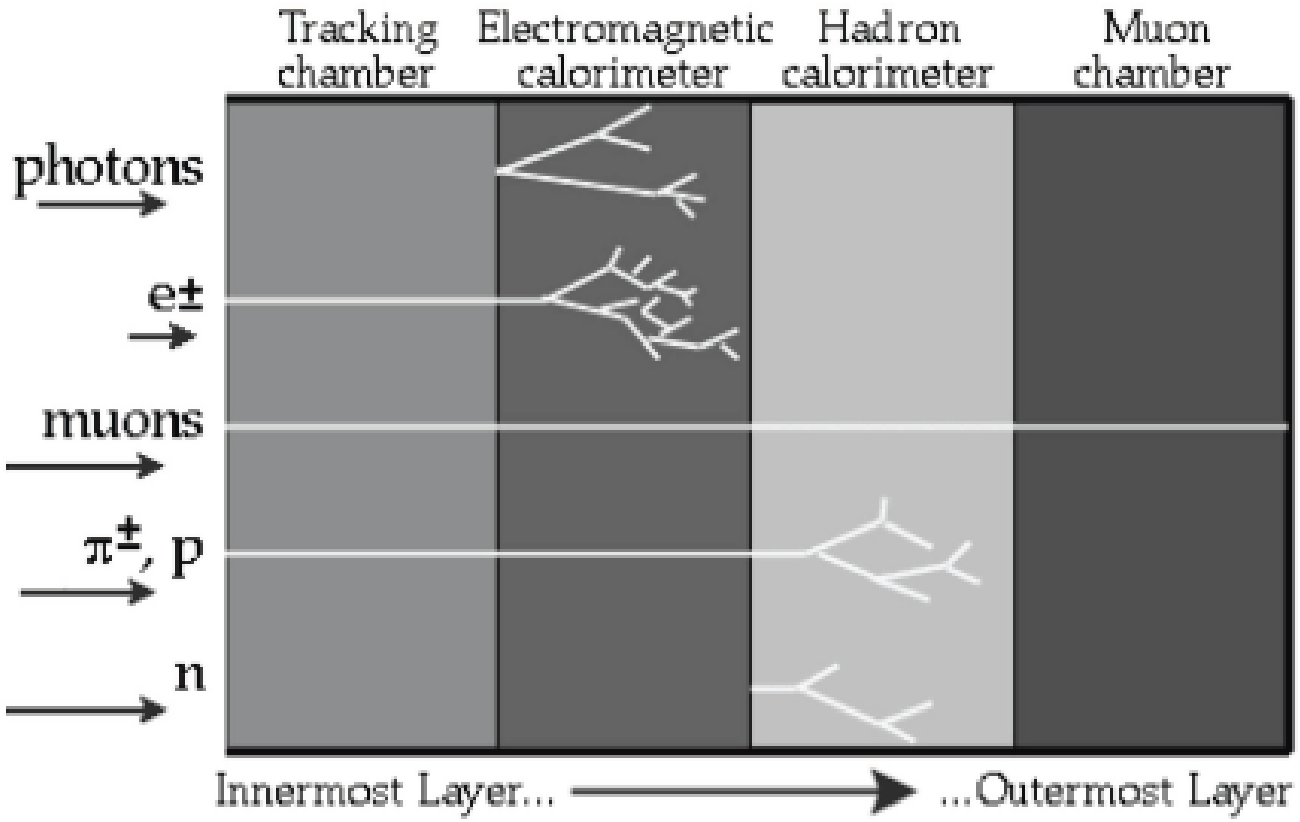
**Poisson process so reso RMS (ΔE or $\sigma_E$) depends on $\sqrt{N}$ → $\sqrt{E}$**

# Sampling Calo

- Typically "sampling" calo like this:

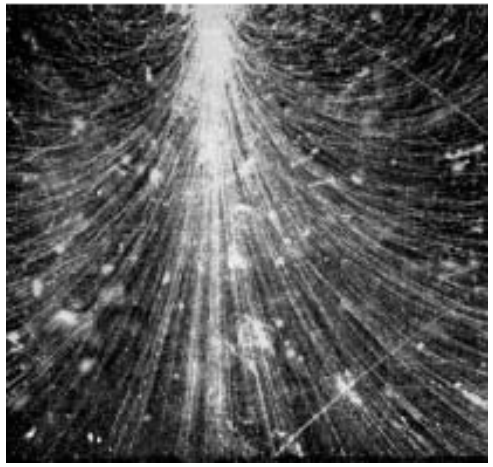


# Calorimeters



# High E e, γ : EM Showering

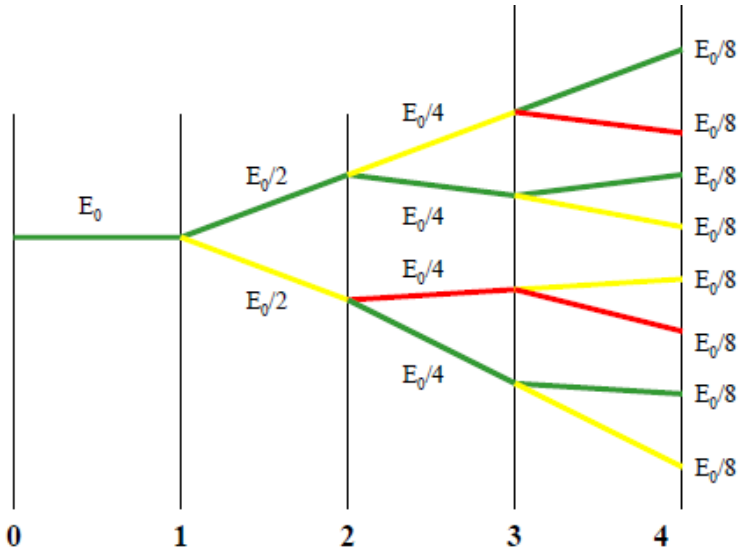
- Alternating:
  - Pair Production γ->e+e-
  - Bremsstrahlung e -> e γ



- Multiplication of particles
  - Even for high energy e,photon detection, measure E by collecting all resulting low E e, photons, (~10000's) through the lower E processes

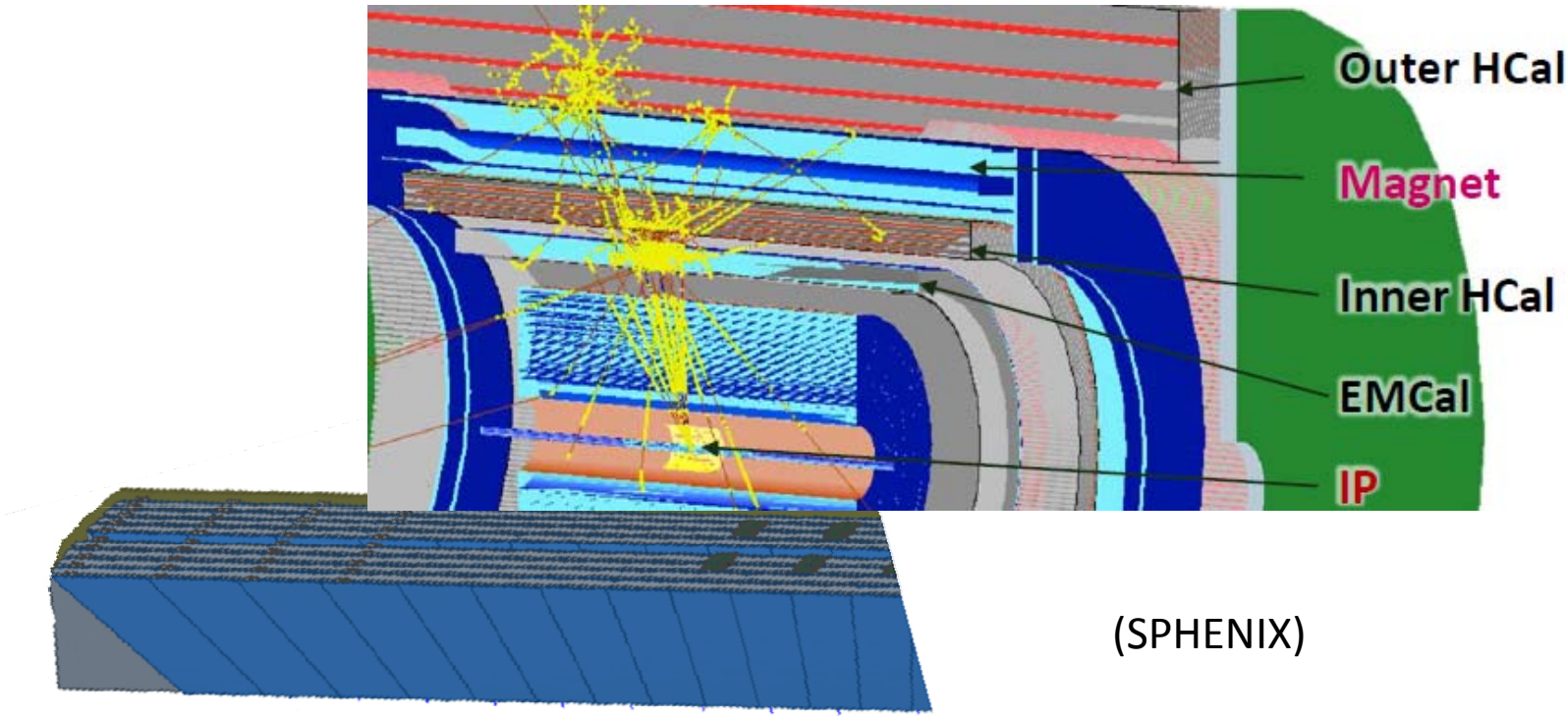
# Hadronic Calos/Showers

- Low energy:  neutral hadrons very hard
- High energy e.g. Neutrons can strike Nuclei, which then emit other high energy hadrons
- Hadronic Showering
  - Only way for neutral particles like n, $K^0_L$
  - charge hadrons detected here too, redundant w/ tracking



# Both Calos

- Use ~cm-size segmentation like one big tracking plane…can't "track" but (single) location and E is enough to do a lot
  - e.g. for EM photons, no bending so we know trajectory already (straight from event "vertex"



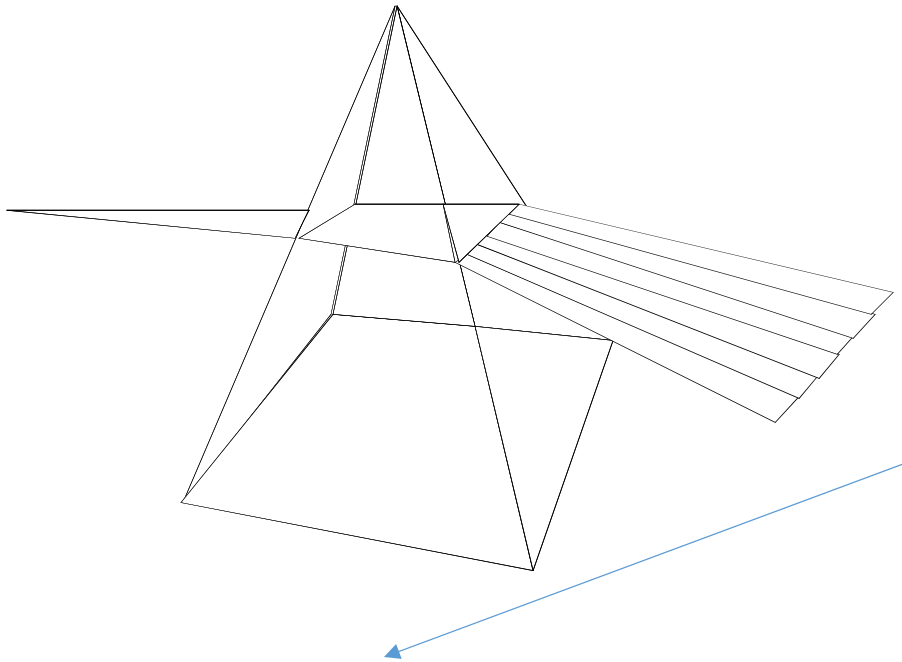

(SPHENIX)

# TRACKING DETECTORS magnetic "spectrometers"

TRACK TRAJECTORIES → BEND
BETTER RESO WHERE BEND IS LARGER→ low MOMENTUM

TYPICALLY Ch tracking detectors placed in Magnet

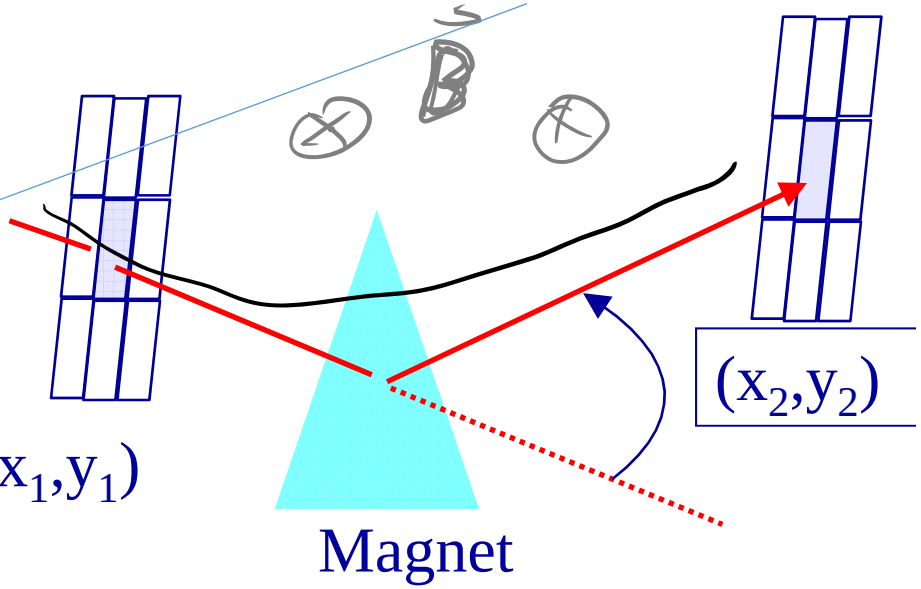
Just as light of different wavelength is bent differently by a prism...



ρ radius of curv. in m

Nature lets us measure the Momentum of a charged particle by seeing how much its path is deflected by a magnet.

$$p(GeV) = 0.3\, B(T)\rho(m)$$




slide from Roche

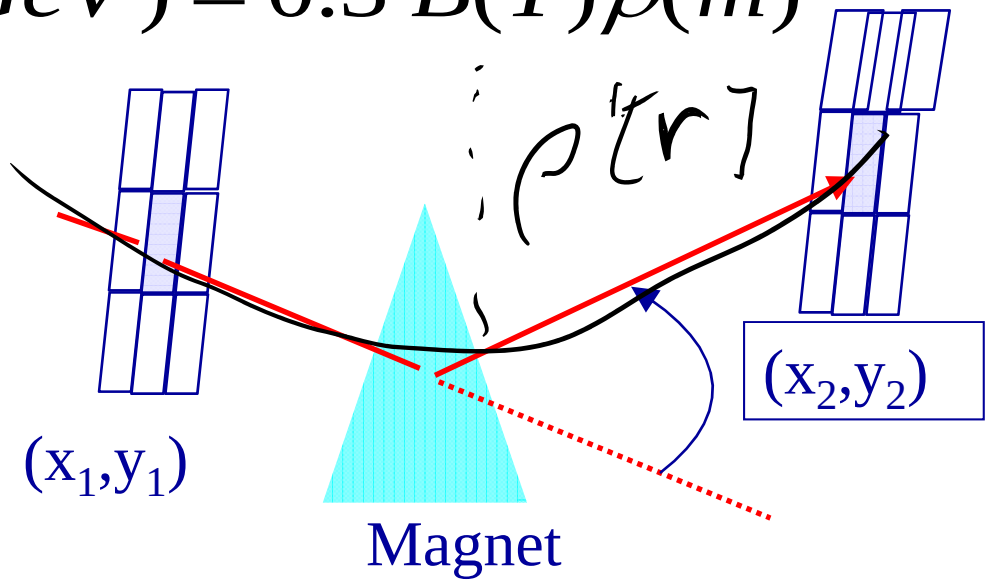
# Magnet + tracking

- Charge tracking detectors ~always (?) used with Magnetic field /Magnet (1T-3T ) field

Two main types of tracking
"whole trajectory"

Planes of space point
Silicon Micro Pixels style most common (segments Δx/y μm)

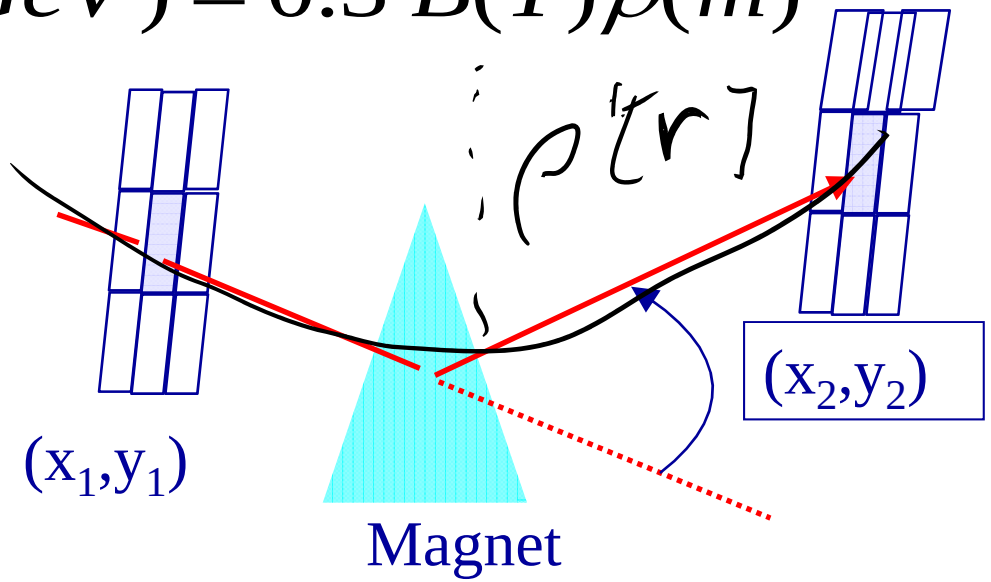
$$p(GeV) = 0.3\, B(T)\rho(m)$$



# Exercise

- Derive the factor of 0.3 (using 1 GeV = 1.6e-10 J, etc, don't forget about c for momentum (GeV/c).
- Also what's the radius of curvature for a 10 GeV charged pion in a 1 T field

$$p(GeV) = 0.3\, B(T)\rho(m)$$



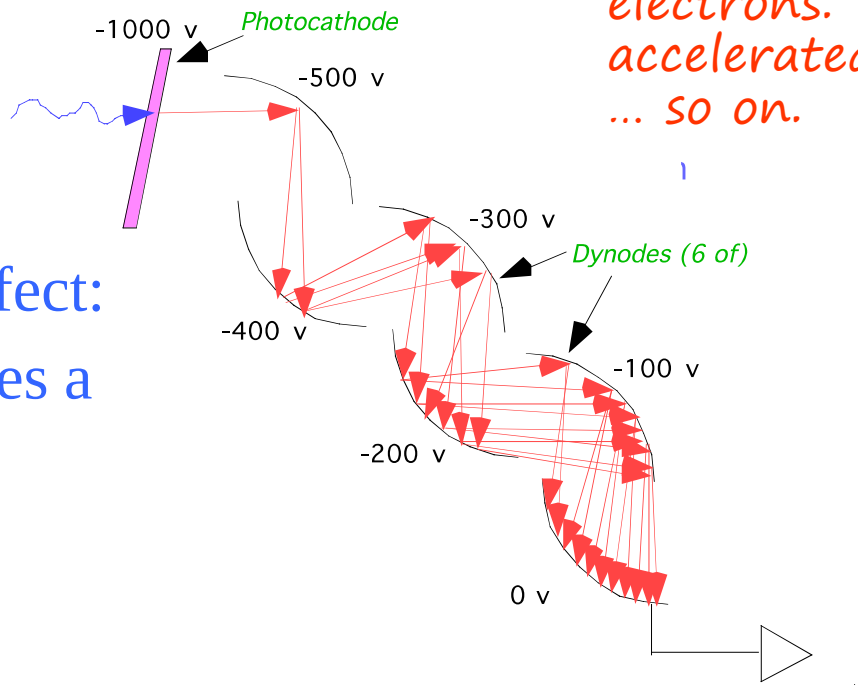
# Particle Detectors: amplification with the photo multiplier

## Secondary Emission

Energetic electrons striking some surfaces can liberate MORE electrons. Those, in turn, can be accelerated onto another surface … so on.

Photoelectric Effect:

A photon liberates a single electron




PMTs are commercially produced and very sensitive.

- One photon --> up to $10^8$ electrons!
- Fast! …down to ~ few x $10^{-9}$ seconds.

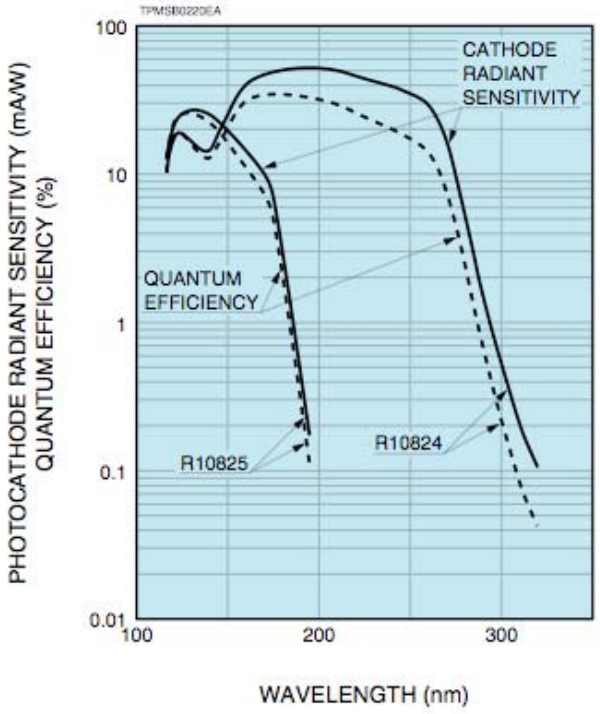


# SPECIFICATIONS

## GENERAL

| Parameter | | R10824 | R10825 | Unit |
|---|---|---|---|---|
| Spectral Response | | 115 to 320 | 115 to 195 | nm |
| Wavelength of Maximum Response | | 200 | 130 | nm |
| Photocathode Material | | Cs-Te | Cs-I | — |
| Window Material | | $MgF_2$ | | — |
| Minimum Effective Area | | 4 × 9.5 | | mm |
| Dynode | Structure | Circular-cage | | — |
| | Number of Stage | 9 | | — |
| | Material | Sb-Cs | | — |
| Direct Interelectrode Capacitances | Anode to Dynode No.9 | Approx. 1.7 | | pF |
| | Anode to All Other Electrodes | Approx. 2.0 | | pF |
| Base | | 11-pin base | | — |
| Weight | | 7.2 | | g |
| Suitable Socket for Base (supplied) | | E678-11U | | — |

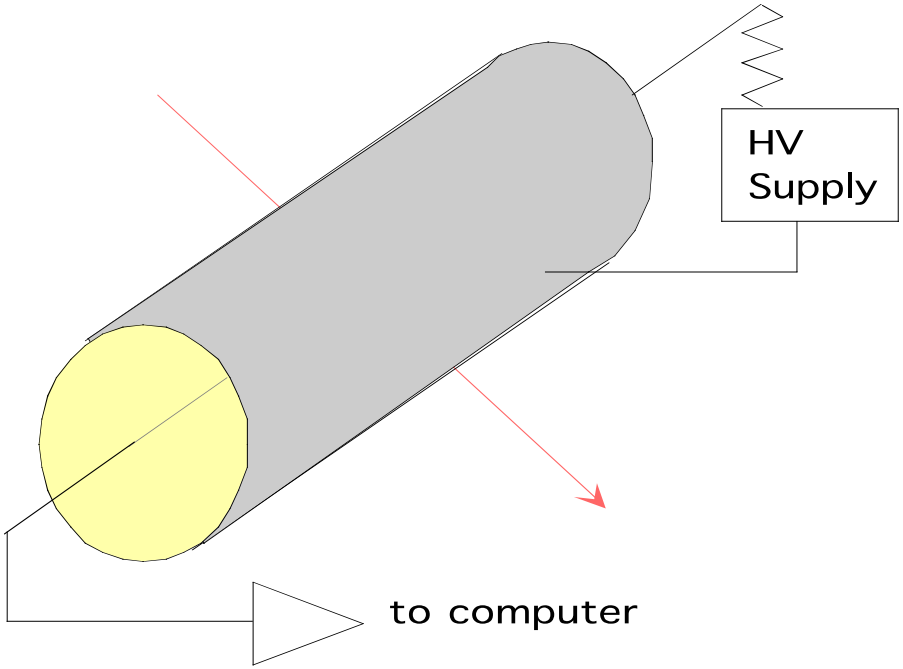
Figure 1: Typical Spectral Response



# Now siPM's

- PMT's are now also being replaced by silicon light detectors primarily siPM's silicon Photon Multipliers
- These work with super large pixilation (micro meters)
- And only require usual silicon bias voltage (Order ~10's V) rather than high V

# Particle Detectors: amplification with the E-Field




$$E(r) = \frac{V}{r \ln(r_c / r_a)}$$

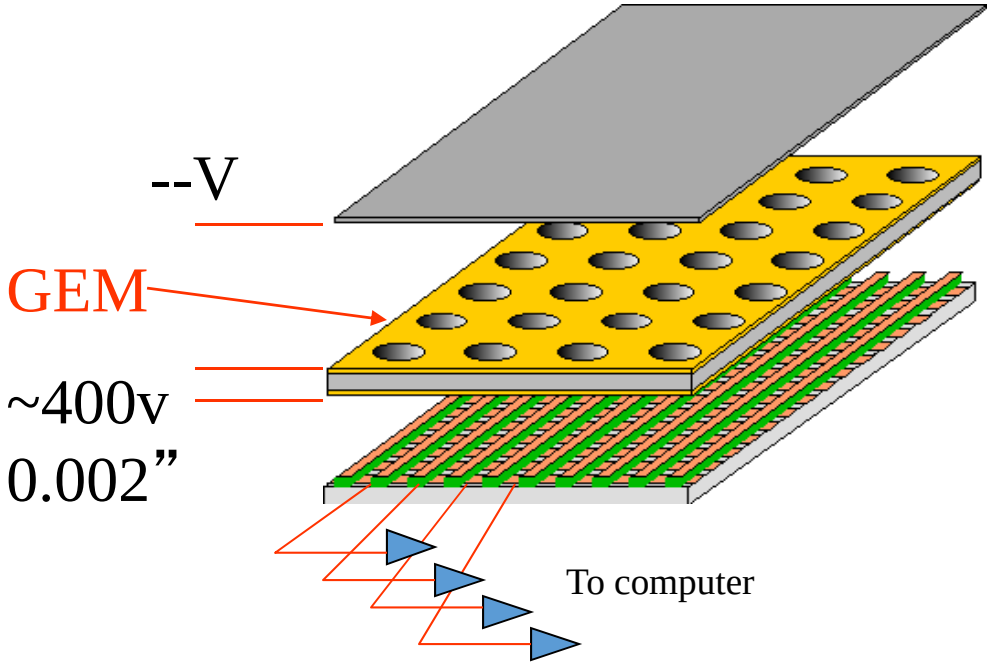
- V is the voltage
- $r_c$ radius of the outer cathode plane
- $r_a$ radius of the inner anode

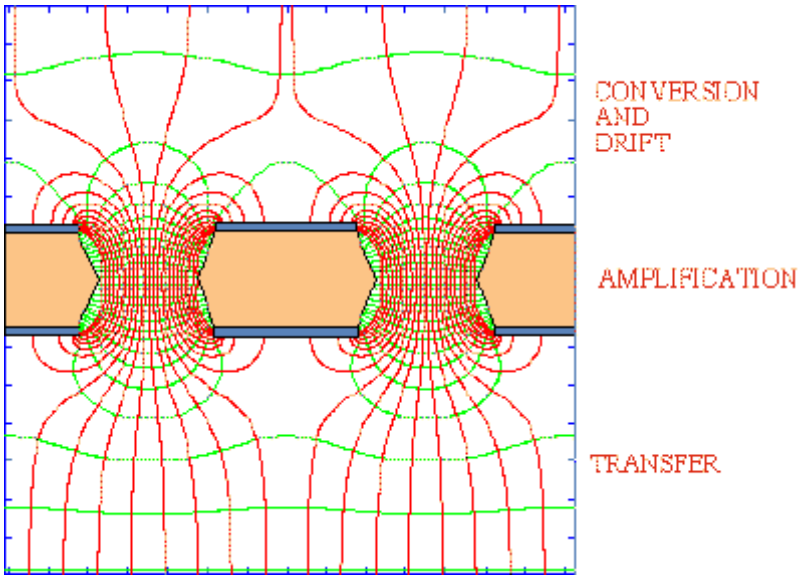
Close to the wire the E field is very Intense, the charged particle is Accelerated and
as a result creates secondary Particles.

# Particle Detectors… *Gas Electron Multiplier (GEM)*

*Gas Ionization and Avalanche Multiplication again, but…*

- … a different way to get an intense electric field,
- … without dealing with fragile tiny wires

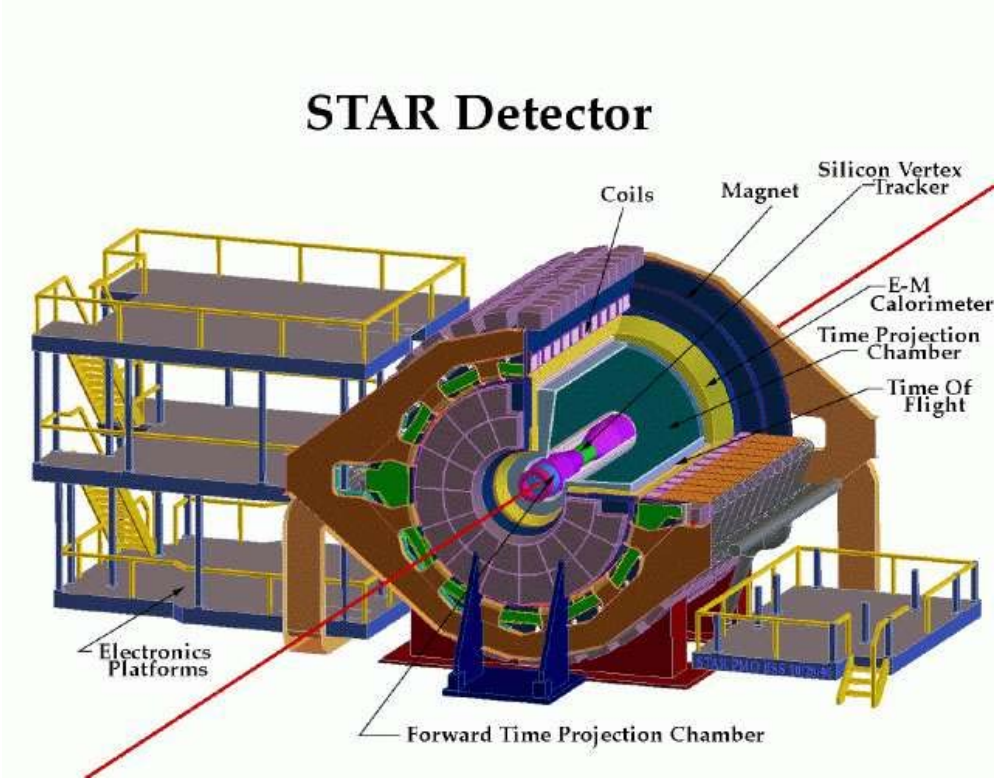



http://gdd.web.cern.ch/GDD/

# Particle detectors

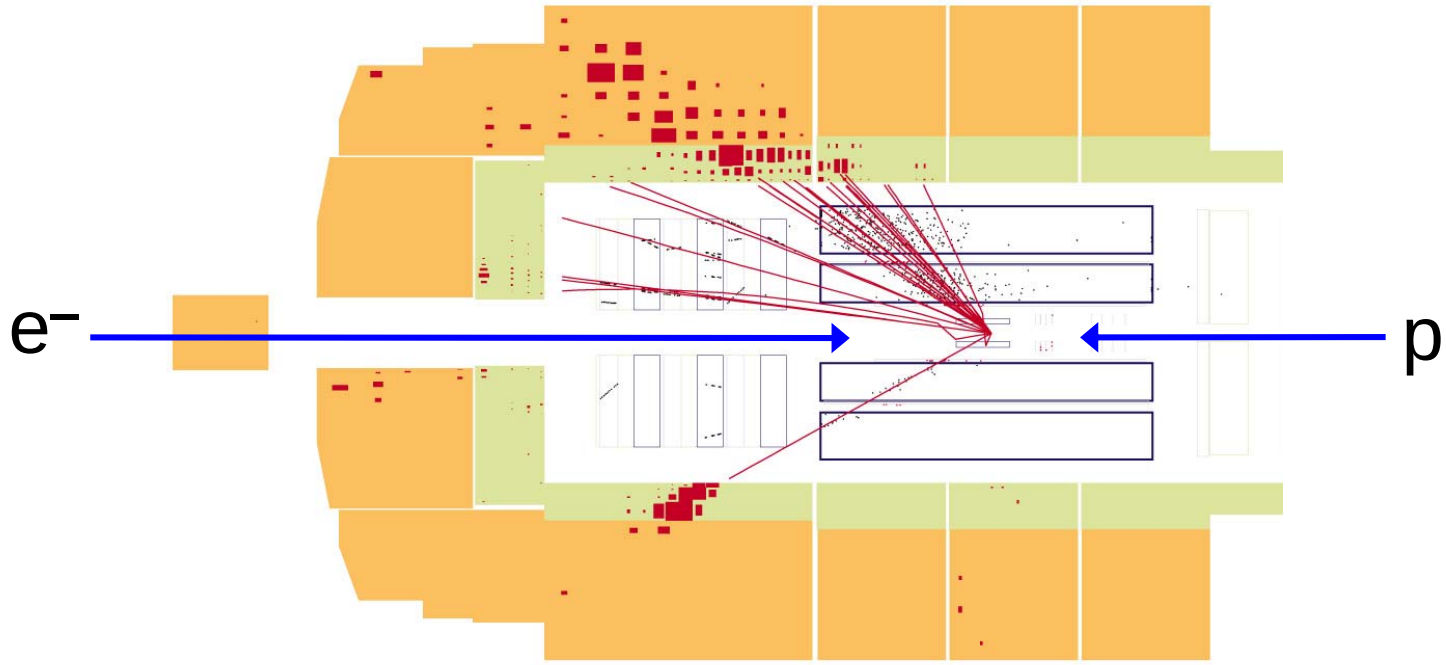
- Avalanche multiplication
- Detectors that see the electrons
  - Wire chambers
  - Time Projection Chamber
- Detectors that see the light
  - Scintillators
  - Cerenkov Detectors
  - Calorimeters



# Particle Physics

**Many slides from here today:**

**Michaelmas Term 2011**
**Prof Mark Thomson**




## Handout 6 : Deep Inelastic Scattering

# Elastic → Inelastic Scattering

★ **Recall:** Elastic scattering (Handout 5)

- Only one independent variable. In Lab. frame express differential cross section in terms of the electron scattering angle (Rosenbluth formula)

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E_1^2 \sin^4\theta/2}\frac{E_3}{E_1}\left(\frac{G_E^2 + \tau G_M^2}{(1+\tau)}\cos^2\frac{\theta}{2} + 2\tau G_M^2 \sin^2\frac{\theta}{2}\right) \qquad \tau = \frac{Q^2}{4M^2}$$

**Note:** here the energy of the scattered electron is determined by the angle.

- In terms of the Lorentz invariant kinematic variables can express this differential cross section in terms of $Q^2$ (Q13 on examples sheet)

$$\frac{d\sigma}{dQ^2} = \frac{4\pi\alpha^2}{Q^4}\left[\frac{G_E^2 + \tau G_M^2}{(1+\tau)}\left(1 - y - \frac{M^2y^2}{Q^2}\right) + \frac{1}{2}y^2 G_M^2\right]$$

which can be written as:

$$\frac{d\sigma}{dQ^2} = \frac{4\pi\alpha^2}{Q^4}\left[f_2(Q^2)\left(1 - y - \frac{M^2y^2}{Q^2}\right) + \frac{1}{2}y^2 f_1(Q^2)\right]$$

G or f called "Form factors"

★ ~~Inelastic scattering~~

- For Deep Inelastic Scattering have two independent variables. Therefore need a double differential cross section

# Size/shape of charge in of proton

First part of elastic

Therefore, in order to account for the extended charge distribution of the proton, the Mott scattering cross section of (7.14) has to be modified to

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Mott}} \rightarrow \frac{\alpha^2}{4E^2 \sin^4(\theta/2)} \cos^2\left(\frac{\theta}{2}\right) |F(\mathbf{q}^2)|^2 . \qquad (7.17)$$

The form factor $F(\mathbf{q}^2)$ is the three-dimensional Fourier transform of the charge distribution. If the wavelength of the virtual photon is large compared to the size of the c

$$F(\mathbf{q}^2) = \int \rho(\mathbf{r}) e^{i\mathbf{q}\cdot\mathbf{r}} \, d^3\mathbf{r}.$$

volume integral. In this cas for a point-like object and th ribution, $F(0)$

# Form factors related to Fourier Transform

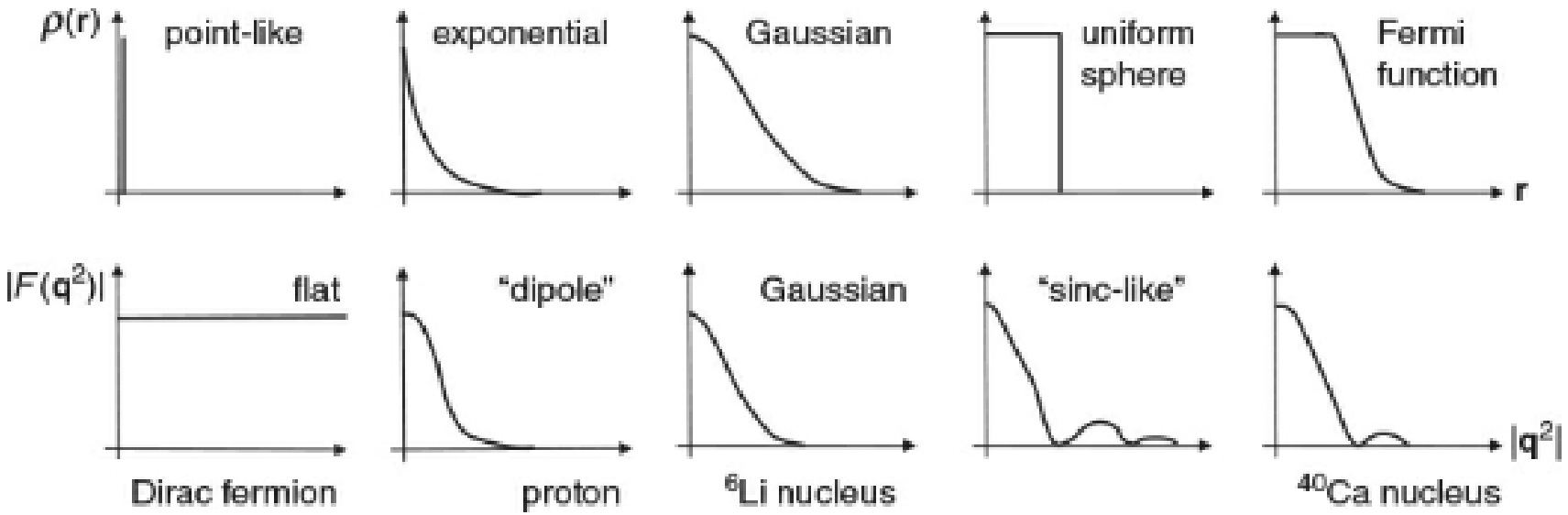



**Fig. 7.5** Possible three-dimensional charge distributions and the corresponding form factors plotted as a function of $\mathbf{q}^2$.

# $e^-$ p Elastic Scattering at Very High $q^2$

★**At high $q^2$ the Rosenbluth expression for elastic scattering becomes**

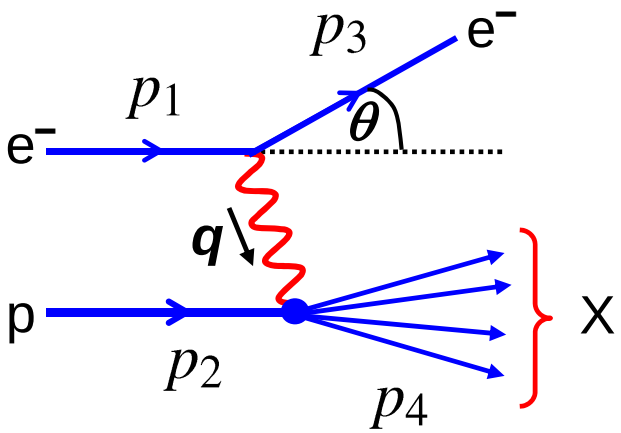
$$\left(\frac{d\sigma}{d\Omega}\right)_{elastic} = \frac{\alpha^2}{4E_1^2 \sin^4\theta/2}\frac{E_3}{E_1}\left(\frac{q^2}{2M^2}G_M^2 \sin^2\frac{\theta}{2}\right) \qquad \tau = -\frac{q^2}{4M^2} \gg 1$$

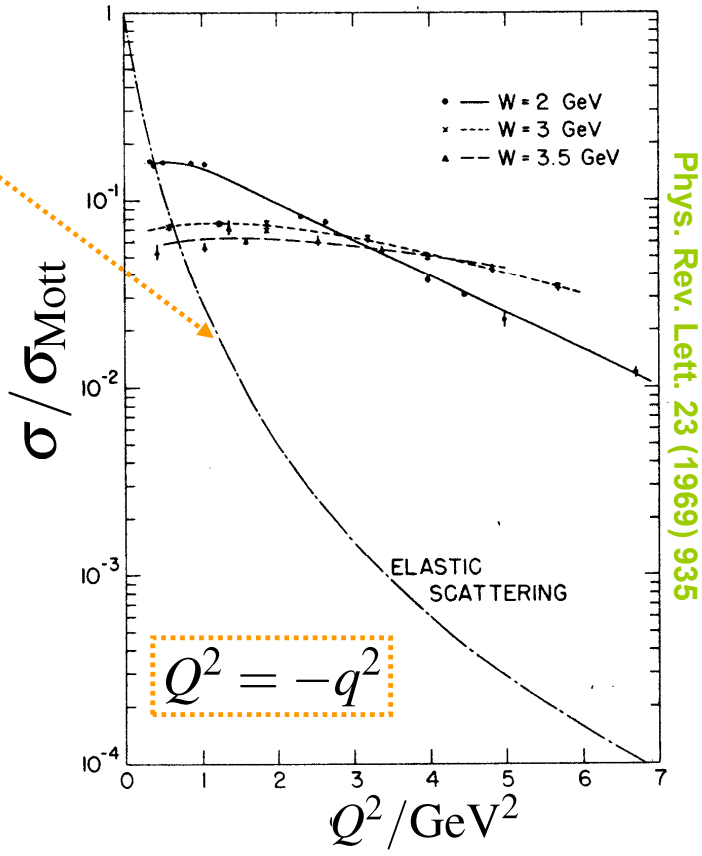
•**From $e^-$ p elastic scattering, the proton magnetic form factor is**

$$G_M(q^2) \approx \frac{1}{(1+q^2/0.71\text{GeV}^2)^2} \Rightarrow G_M(q^2) \propto q^{-4} \quad \text{at high } q^2$$

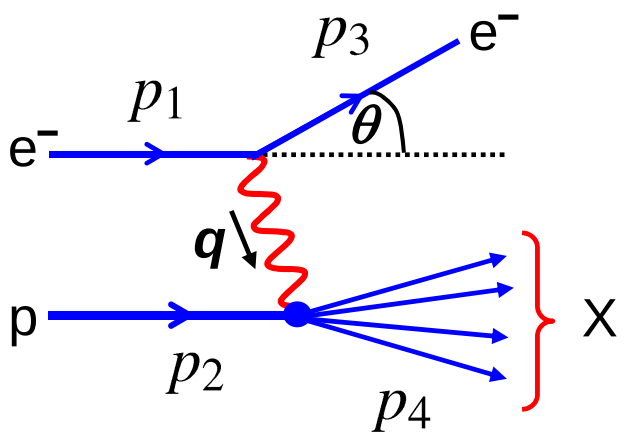
$$\Rightarrow \left(\frac{d\sigma}{d\Omega}\right)_{elastic} \propto q^{-6}$$

•**Due to the finite proton size, elastic scattering at high $q^2$ is unlikely and inelastic reactions where the proton breaks up dominate.**



# Kinematics of Inelastic Scattering




- For inelastic scattering the mass of the final state hadronic system is no longer the proton mass, ***M***
- The final state hadronic system must contain at least one **baryon** which implies the final state invariant mass $M_X > M$

$$M_X^2 = p_4^2 = (E_4^2 - |\vec{p}_4|^2)$$

✶ For inelastic scattering introduce four new kinematic variables: $x, y, \nu, Q^2$

★ **Define:**

$$x \equiv \frac{Q^2}{2p_2.q}$$

**Bjorken x** (Lorentz Invariant)

where $Q^2 \equiv -q^2$ $\quad Q^2 > 0$

- Here $M_X^2 = p_4^2 = (q + p_2)^2 = -Q^2 + 2p_2.q + M^2$

$\Rightarrow$ $Q^2 = 2p_2.q + M^2 - M_X^2$ $\Rightarrow$ $Q^2 \leq 2p_2.q$

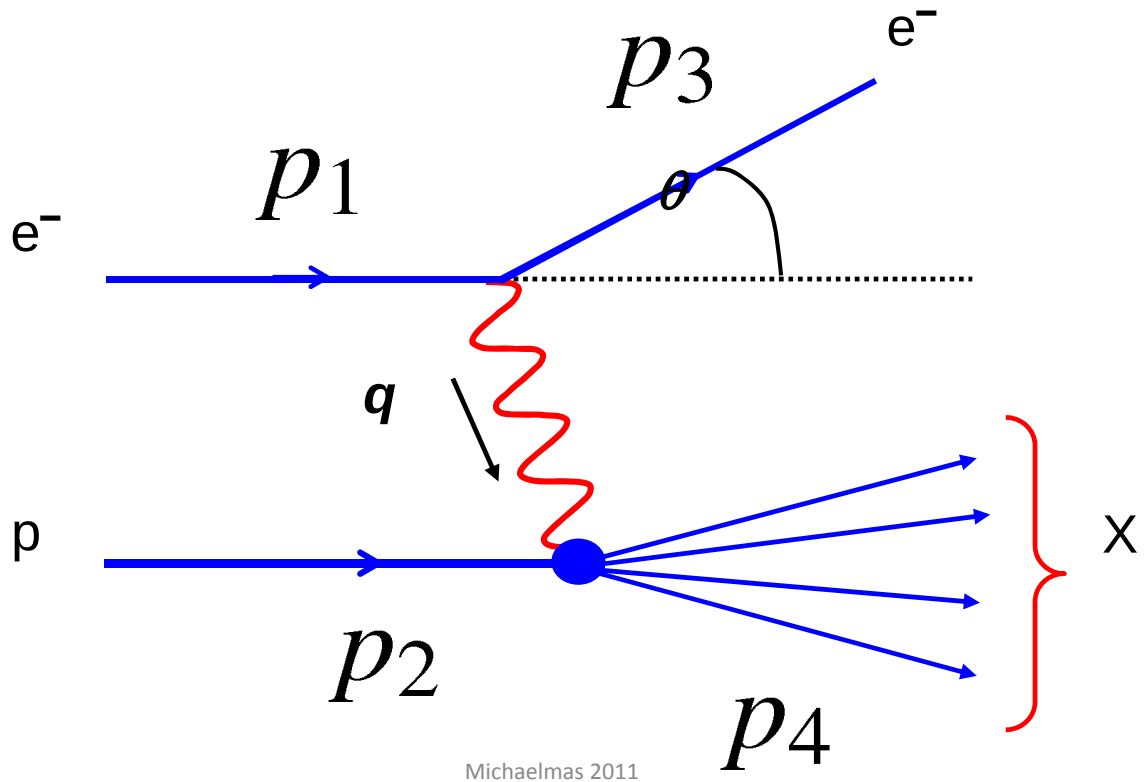
**Note: in many text books *W* is often used in place of $M_X$**

hence $0 < x < 1$ **inelastic** $\quad x = 1$ **elastic**

**Proton intact** $M_X = M$

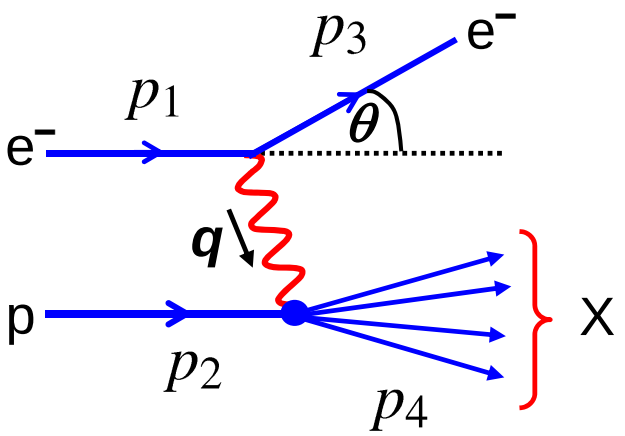
# Also note, t-channel ONLY

- This implies the s is not as important, t more, but also…
- For example for production of NEW particle amongst or as X what's relavant is the microscopic details of the bottom vertex
  - t = only one input
  - The other input is from proton "blob" → related to little x
  - "how inelastic" → directly related to how much mom quark/glu have




Imagine What mX's are possible ? depends on q = (sqrt(t)) & contribution from proton→ "x"

★ **Define:** $y \equiv \dfrac{p_2 . q}{p_2 . p_1}$ (Lorentz Invariant)




•**In the Lab. Frame:**

$p_1 = (E_1, 0, 0, E_1) \quad p_2 = (M, 0, 0, 0)$

$q = (E_1 - E_3, \vec{p}_1 - \vec{p}_3)$

$$\rightarrow \quad y = \frac{M(E_1 - E_3)}{ME_1} = 1 - \frac{E_3}{E_1}$$

So ***y*** **is the fractional energy loss of the incoming particle**

$$0 < y < 1$$

•**In the C.o.M. Frame (neglecting the electron and proton masses):**

$$p_1 = (E, 0, 0, E); \quad p_2 = (E, 0, 0, -E); \quad p_3 = (E, E\sin\theta^*, 0, E\cos\theta^*)$$

$$\rightarrow \quad y = \tfrac{1}{2}(1 - \cos\theta^*) \qquad \text{for} \quad E \gg M$$

★ **Finally Define:** $\nu \equiv \dfrac{p_2 . q}{M}$ (Lorentz Invariant)

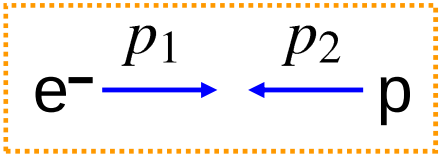
•**In the Lab. Frame:** $\nu = E_1 - E_3$

$\nu$ **is the energy lost by the incoming particle**

**y has nothing to do with rapidity**
**y and nu are not very important out of the other variables**

# Relationships between Kinematic Variables

- Can rewrite the new kinematic variables in terms of the squared centre-of-mass energy, **s**, for the electron-proton collision




$$s = (p_1 + p_2)^2 = p_1^2 + p_2^2 + 2p_1.p_2 = 2p_1.p_2 + M^2 + \cancel{m_e^2}$$

Neglect mass of electron

$$2p_1.p_2 = s - M^2$$

- For a fixed centre-of-mass energy, it can then be shown that the four kinematic variables

$$Q^2 \equiv -q^2 \qquad x \equiv \frac{Q^2}{2p_2.q} \qquad y \equiv \frac{p_2.q}{p_2.p_1} \qquad \nu \equiv \frac{p_2.q}{M}$$

are not independent.

- i.e. the scaling variables $x$ and $y$ can be expressed as

$$x = \frac{Q^2}{2M\nu} \qquad y = \frac{2M}{s - M^2}\nu$$

Note the simple relationship between ***y*** and ν

and $$xy = \frac{Q^2}{s - M^2} \quad \Rightarrow \quad Q^2 = (s - M^2)xy$$

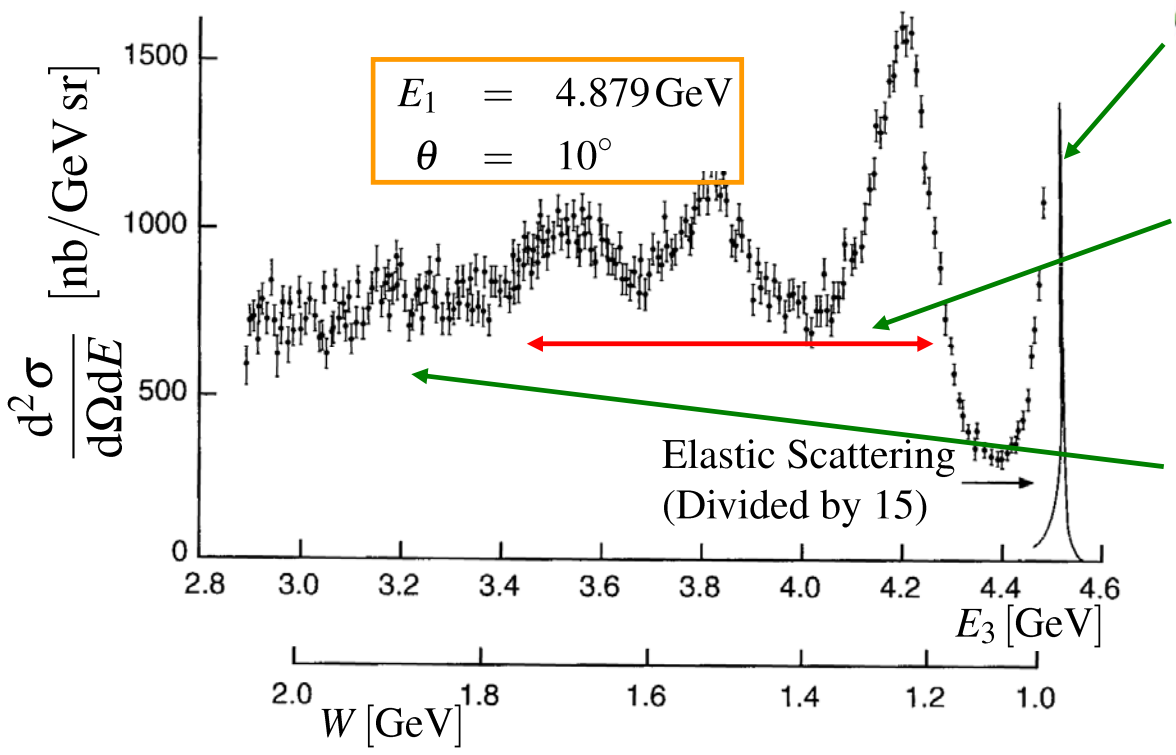
- For a fixed centre of mass energy, the interaction kinematics are completely defined by any **two** of the above kinematic variables (except ***y*** and ν)

- For elastic scattering $(x = 1)$ there is only one independent variable. As we saw previously if you measure electron scattering angle know everything else.

# Inelastic Scattering

**Example:** Scattering of 4.879 GeV electrons from protons at rest

- Place detector at $10^\circ$ to beam and measure the energies of scattered $e^-$
- Kinematics fully determined from the electron energy and angle !
- e.g. for this energy and angle : the invariant mass of the final state hadronic system $W^2 = M_X^2 = 10.06 - 2.03E_3$ (try and show this)




- **Elastic Scattering**
  proton remains intact
  $W = M$
- **Inelastic Scattering**
  produce "excited states" of proton e.g. $\Delta^+(1232)$
  $W = M_\Delta$
- **Deep Inelastic Scattering**
  proton breaks up resulting in a many particle final state

**DIS = large $W$**

# Griff

- Griff in direction of W

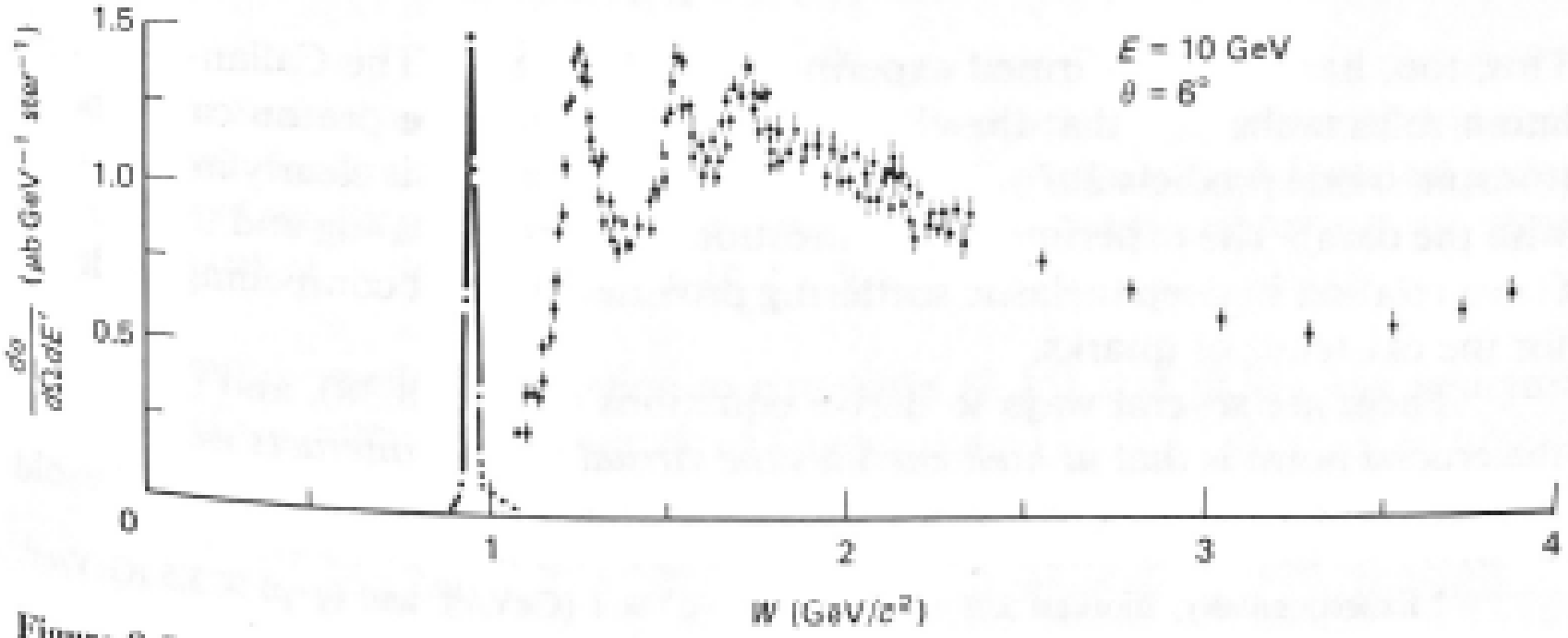
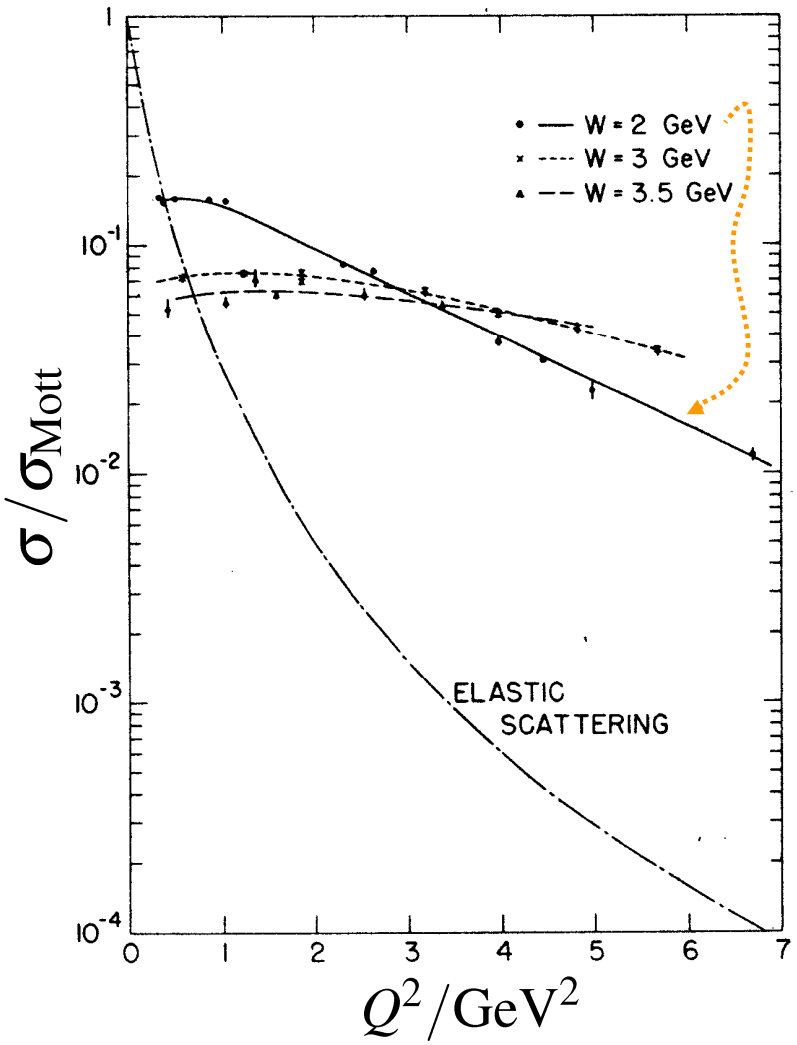



Figure 8.5 Cross section for inclusive inelastic electron-proton scattering, as a function of "missing

# Inelastic Cross Sections

- Repeat experiments at different angles/beam energies and determine $q^2$ dependence of elastic and inelastic cross-sections




M.Breidenbach et al., Phys. Rev. Lett. 23 (1969) 935

- Elastic scattering falls of rapidly with $q^2$ due to the proton not being point-like (i.e. form factors)
- Inelastic scattering cross sections only weakly dependent on $q^2$
- Deep Inelastic scattering cross sections almost independent of $q^2$ !

i.e. "Form factor" → 1

→ **Scattering from point-like objects within the proton !**

# Deep Inelastic Scattering

★ It can be shown that **the most general Lorentz Invariant expression** for $e^- p \to e^- X$ inelastic scattering (via a single exchanged photon is):

$$\frac{d^2\sigma}{dx dQ^2} = \frac{4\pi\alpha^2}{Q^4}\left[\left(1 - y - \frac{M^2 y^2}{Q^2}\right)\frac{F_2(x,Q^2)}{x} + y^2 F_1(x,Q^2)\right] \quad (1)$$

**INELASTIC SCATTERING**

**c.f.**

$$\frac{d\sigma}{dQ^2} = \frac{4\pi\alpha^2}{Q^4}\left[\left(1 - y - \frac{M^2 y^2}{Q^2}\right) f_2(Q^2) + \frac{1}{2} y^2 f_1(Q^2)\right]$$

**ELASTIC SCATTERING**

We will soon see how this connects to the quark model of the proton

• **NOTE:** The form factors have been replaced by the **STRUCTURE FUNCTIONS**

$$F_1(x,Q^2) \quad \text{and} \quad F_2(x,Q^2)$$

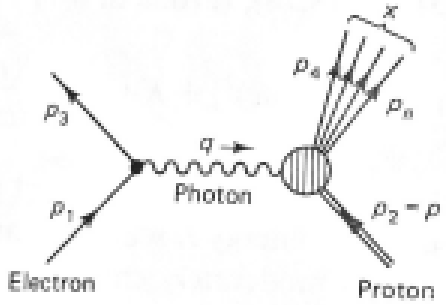
which are a function of ***x*** and $Q^2$: can not be interpreted as the Fourier transforms of the charge and magnetic moment distributions. We shall soon see that they describe the **momentum distribution** of the quarks within the proton

★ In the limit of high energy (or more correctly $Q^2 \gg M^2 y^2$ ) eqn. (1) becomes:

$$\frac{d^2\sigma}{dx dQ^2} = \frac{4\pi\alpha^2}{Q^4}\left[(1 - y)\frac{F_2(x,Q^2)}{x} + y^2 F_1(x,Q^2)\right] \quad (2)$$

# Griff

- New Cross section formula from the Golden Rule needed



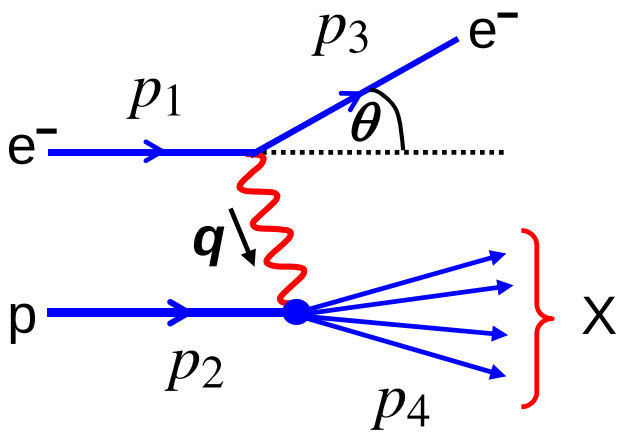

Again, the blob veils our ignorance about the photon-proton vertex; the extra outgoing lines represent the hadronic "shrapnel" from the exploded proton. As before, the *electron* vertex is unaffected by all the excitement at the proton end, and so the (spin-averaged) amplitude, for a given final state $X$, takes the form

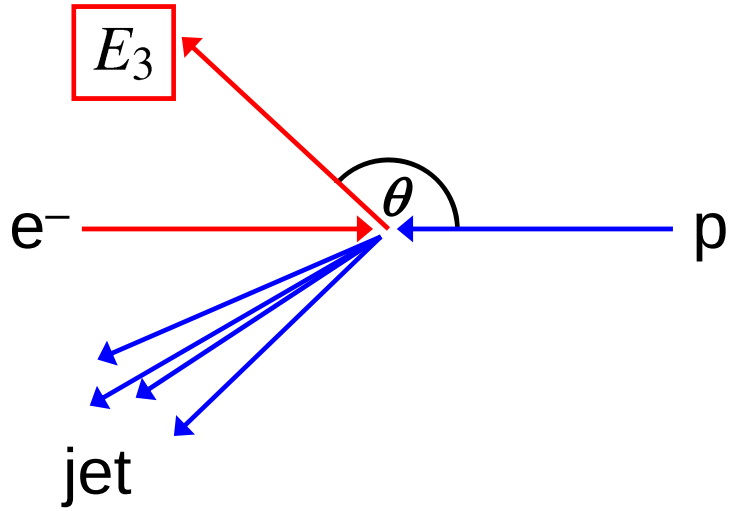
$$\langle |\mathcal{M}|^2 \rangle = \frac{g_e^4}{q^4} L^{\mu\nu}_{\text{electron}} K_{\mu\nu}(X) \tag{8.24}$$

where $K_{\mu\nu}$ is some (unknown) quantity describing the subprocess $\gamma + p \to X$; it depends on $q = (p_1 - p_3)$, $p = p_2$, and the various outgoing momenta $p_4, p_5, \ldots, p_n$. The scattering cross section is determined by the "Golden Rule" [eq. (6.34)]:

$$d\sigma = \frac{\hbar^2 \langle |\mathcal{M}|^2 \rangle}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left( \frac{c\, d^3\mathbf{p}_3}{(2\pi)^3 2E_3} \right) \left( \frac{c\, d^3\mathbf{p}_4}{(2\pi)^3 2E_4} \right) \cdots \left( \frac{c\, d^3\mathbf{p}_n}{(2\pi)^3 2E_n} \right) \times (2\pi)^4\, \delta^4(p_1 + p_2 - p_3 - p_4 - \cdots - p_n) \tag{8.25}$$

- **In the Lab. frame it is convenient to express the cross section in terms of the angle, $\theta$ , and energy, $E_3$, of the scattered electron – experimentally well measured.**




$$Q^2 = 4E_1E_3\sin^2\theta/2; \quad x = \frac{Q^2}{2M(E_1-E_3)}; \quad y = 1-\frac{E_3}{E_1}; \quad \nu = E_1 - E_3$$

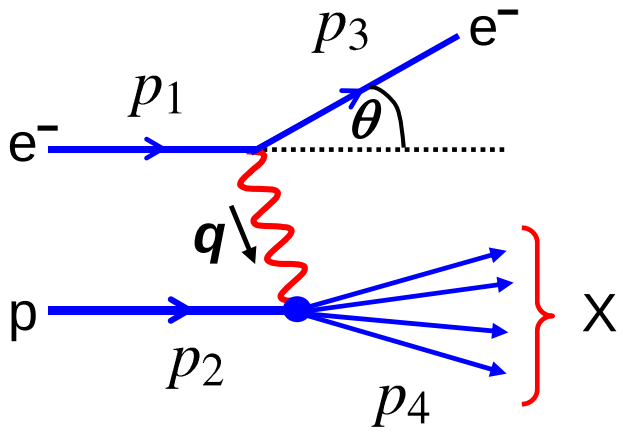
- **In the Lab. frame, Equation (2) becomes:** **(see examples sheet Q13)**

$$\frac{d^2\sigma}{dE_3 d\Omega} = \frac{\alpha^2}{4E_1^2\sin^4\theta/2}\left[\frac{1}{\nu}F_2(x,Q^2)\cos^2\frac{\theta}{2} + \frac{2}{M}F_1(x,Q^2)\sin^2\frac{\theta}{2}\right] \quad (3)$$

**Electromagnetic Structure Function**

**Pure Magnetic Structure Function**

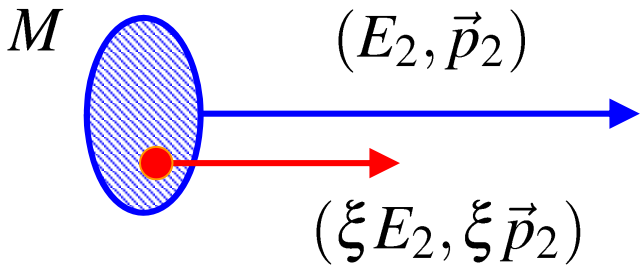
# Now let's understand x better

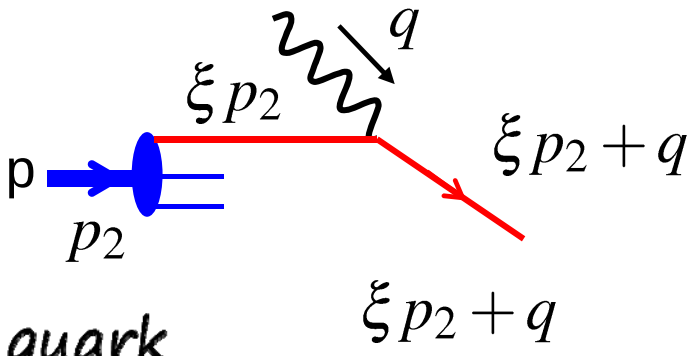



mX possible is from the "s" of this vertex:

$$(\xi p_2 + q)^2$$

- Let the quark carry a fraction $\xi$ of the proton's four-momentum.

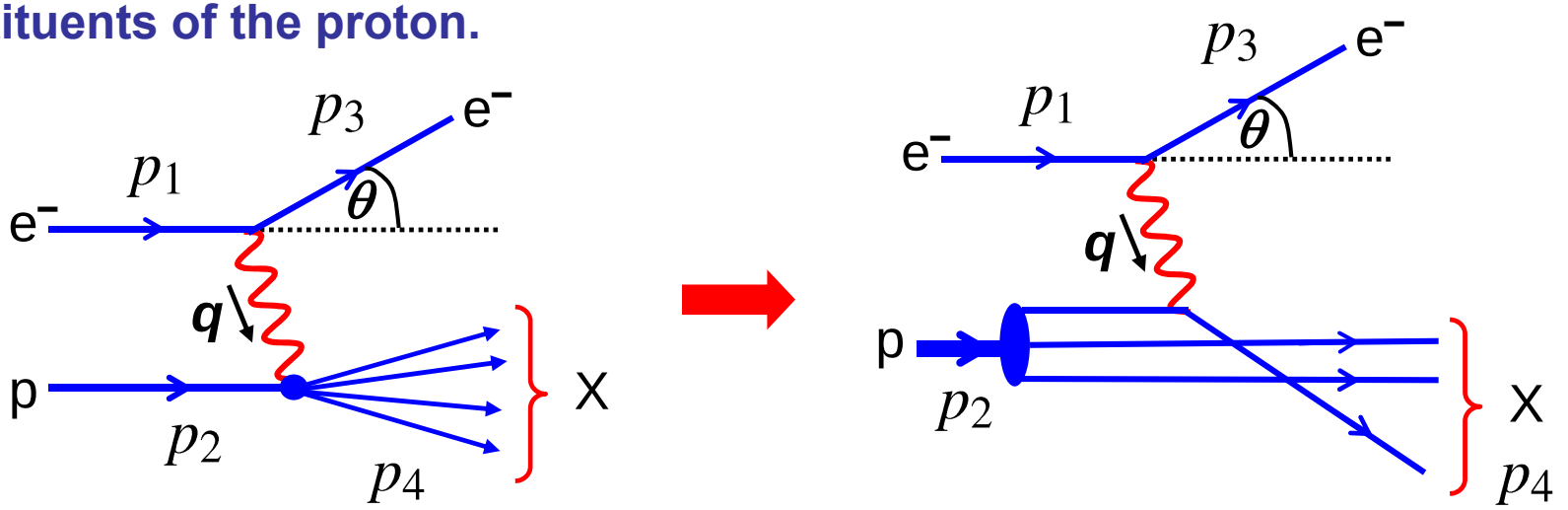



- BUT IF incoming piece of proton is quark and it remains a quark, then mX = mq ~= 0
In this case we can exactly determine ξ

# The Quark-Parton Model

- Before quarks and gluons were generally accepted Feynman proposed that the proton was made up of point-like constituents "partons"
- Both Bjorken Scaling and the Callan-Gross relationship can be explained by assuming that Deep Inelastic Scattering is dominated by the scattering of a single virtual photon from point-like spin-half constituents of the proton.



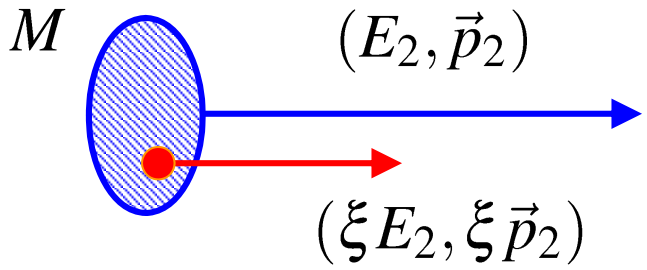

**Scattering from a proton with structure functions** → **Scattering from a point-like quark within the proton**

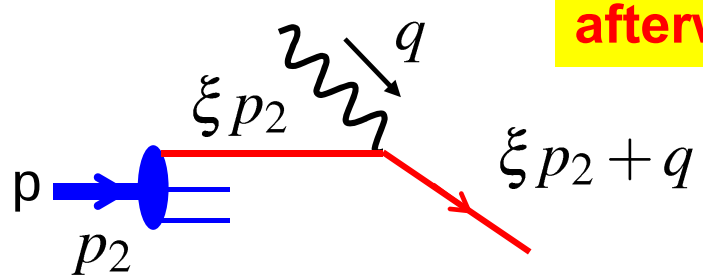
★ How do these two pictures of the interaction relate to each other?

- In the parton model the basic interaction is **ELASTIC** scattering from a "quasi-free" spin-½ quark in the proton, i.e. treat the quark as a free particle!
- The parton model is most easily formulated in a frame where the proton has very high energy, often referred to as the "infinite momentum frame", where we can neglect the proton mass and $p_2 = (E_2, 0, 0, E_2)$
- In this frame can also neglect the mass of the quark and any momentum transverse to the direction of the proton.

Also ignoring q→ hadrons afterwards

- Let the quark carry a fraction $\xi$ of the proton's four-momentum.




- After the interaction the struck quark's four-momentum is $\xi p_2 + q$

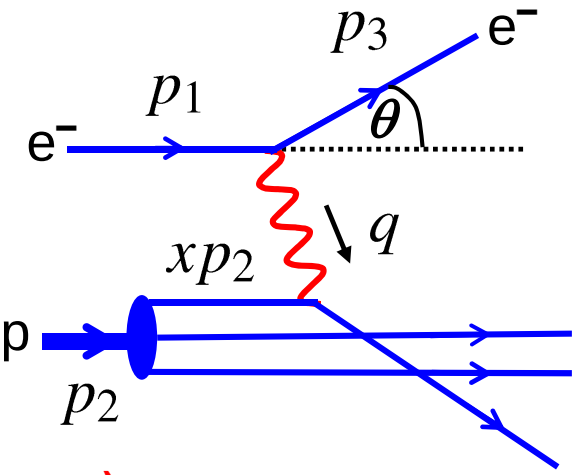
$$(\xi p_2 + q)^2 = m_q^2 \approx 0 \quad \Rightarrow \quad \xi^2 p_2^2 + q^2 + 2\xi p_2 . q = 0 \qquad (\xi^2 p_2^2 = m_q^2 \approx 0)$$

$$\Rightarrow \quad \xi = \frac{Q^2}{2 p_2 . q} = x$$

**Bjorken *x* can be identified as the fraction of the proton momentum carried by the struck quark (in a frame where the proton has very high energy)**

- **In terms of the proton momentum**

$$s = (p_1 + p_2)^2 \simeq 2p_1.p_2 \qquad y = \frac{p_2.q}{p_2.p_1} \qquad x = \frac{Q^2}{2p_2.q}$$




- **But for the underlying quark interaction**

$$s^q = (p_1 + xp_2)^2 = 2xp_1.p_2 = xs$$

$$y_q = \frac{p_q.q}{p_q.p_1} = \frac{xp_2.q}{xp_2.p_1} = y$$

$$x_q = 1$$ **(elastic, i.e. assume quark does not break up )**

- **Previously derived the Lorentz Invariant cross section for $e^-\mu^- \to e^-\mu^-$ elastic scattering in the ultra-relativistic limit (handout 4 + Q10 on examples sheet).**

**Now apply this to** $e^-q \to e^-q$

$$\frac{d\sigma}{dq^2} = \frac{2\pi\alpha^2 e_q^2}{q^4}\left[1 + \left(1 + \frac{q^2}{s_q}\right)^2\right]$$

**$e_q$ is quark charge, i.e.** $e_u = +2/3; \quad e_d = -1/3$

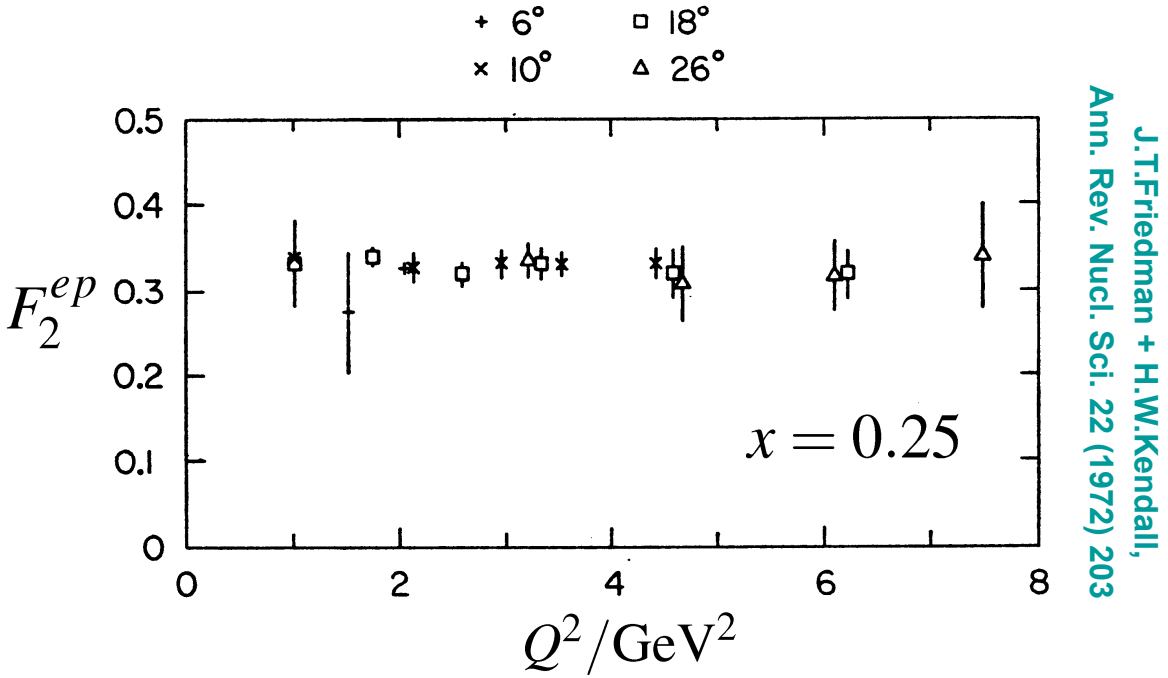
- **Using** $-q^2 = Q^2 = (s_q - m^2)x_q y_q$ ➡ $\frac{q^2}{s_q} = -y_q = -y$

$$\frac{d\sigma}{dQ^2} = \frac{2\pi\alpha^2 e_q^2}{Q^4}\left[1 + (1-y)^2\right]$$

# Measuring the Structure Functions

★ To determine $F_1(x, Q^2)$ and $F_2(x, Q^2)$ for a given $x$ and $Q^2$ need measurements of the differential cross section at several different scattering angles and incoming electron beam energies (see Q13 on examples sheet)

**Example:** electron-proton scattering $F_2$ vs. $Q^2$ at fixed $x$




J.T.Friedman + H.W.Kendall, Ann. Rev. Nucl. Sci. 22 (1972) 203

♦ Experimentally it is observed that both $F_1$ and $F_2$ are (almost) independent of $Q^2$

# FLAT BECAUSE Elastic POINTLIKE

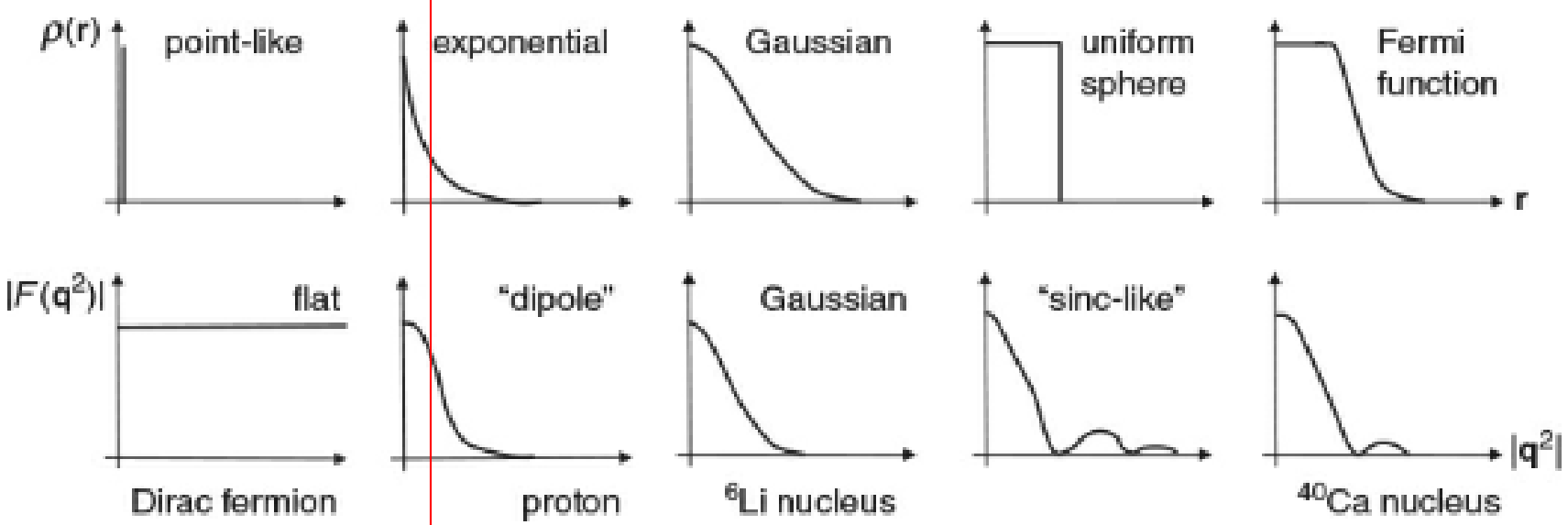



**Fig. 7.5** Possible three-dimensional charge distributions and the corresponding form factors plotted as a function of $\mathbf{q}^2$.

# Griff

$$\frac{d\sigma}{dE'\,d\Omega} = \left(\frac{\alpha\hbar}{2E\sin^2(\theta/2)}\right)^2 [2W_1 \sin^2(\theta/2) + W_2 \cos^2(\theta/2)] \qquad (8.33)$$

for the [illegible] equation (8.21) and $x$ is fixed ($x = 1$). In a formal sense, elastic scattering is a special case of inelastic scattering, in which we impose an extra constraint ($p_{\text{fin}}^2 = M^2c^2$) on the outgoing hadron momenta. It should be possible, therefore, to obtain the Rosenbluth formula from equation (8.33), by appropriate choice of the $W$'s. You can check for yourself that

$$W_{1,2}(q^2, x) = -\frac{K_{1,2}(q^2)}{2Mq^2}\,\delta(x - 1) \qquad (8.35)$$

No K() needed because assume partons ARE point particles

electron-muon results. According to equations (8.35) and (8.36), the structure functions for scattering off a quark of flavor $i$ are

$$W_1^i = \frac{Q_i^2}{2m_i}\,\delta(x_i - 1), \qquad W_2^i = -\frac{2m_i c^2 Q_i^2}{q^2}\,\delta(x_i - 1) \qquad (8.40)$$

# Bjorken Scaling and the Callan-Gross Relation

★The near (see later) independence of the structure functions on $Q^2$ is known as **Bjorken Scaling**, i.e.

$$F_1(x,Q^2) \rightarrow F_1(x) \qquad F_2(x,Q^2) \rightarrow F_2(x)$$

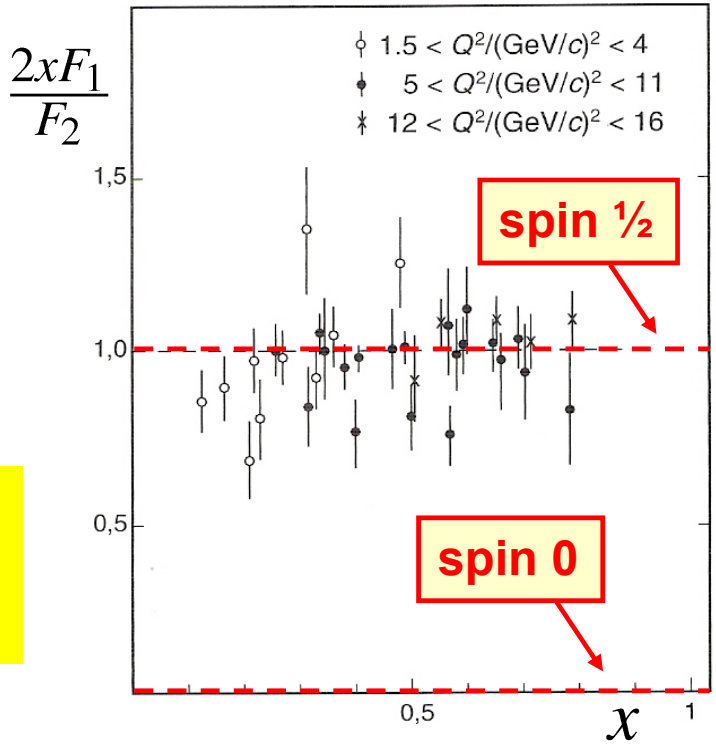
•It is strongly suggestive of scattering from **point-like** constituents within the proton

★It is also observed that $F_1(x)$ and $F_2(x)$ are not independent but satisfy the **Callan-Gross relation**

$$F_2(x) = 2xF_1(x)$$

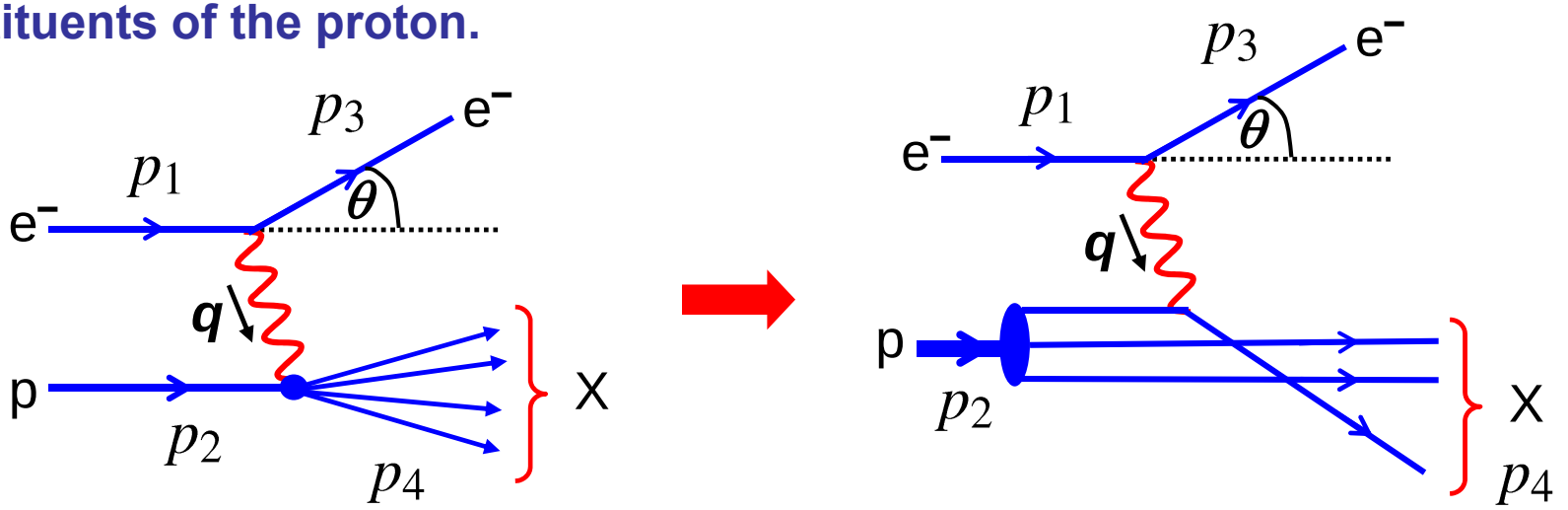
•As we shall soon see this is exactly what is expected for scattering from **spin-half** quarks.

•**Note** if quarks were spin zero particles we would expect the purely magnetic structure function to be zero, i.e. $F_1(x) = 0$



# The Quark-Parton Model

- Before quarks and gluons were generally accepted Feynman proposed that the proton was made up of point-like constituents "partons"
- Both Bjorken Scaling and the Callan-Gross relationship can be explained by assuming that Deep Inelastic Scattering is dominated by the scattering of a single virtual photon from point-like spin-half constituents of the proton.




**Scattering from a proton with structure functions** → **Scattering from a point-like quark within the proton**

★ How do these two pictures of the interaction relate to each other?

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# AT HIGH ENERGY!!!

- Let the quark carry a fraction $\xi$ of the proton's four-momentum.



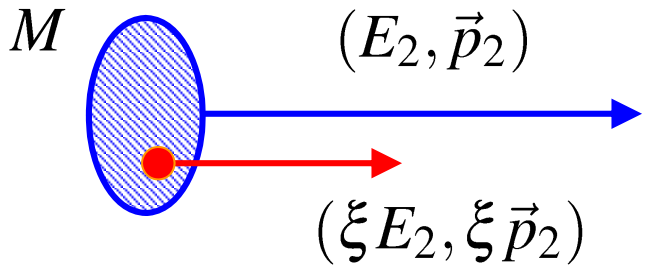

- After the interaction the struck quark's four-momentum is $\xi p_2 + q$

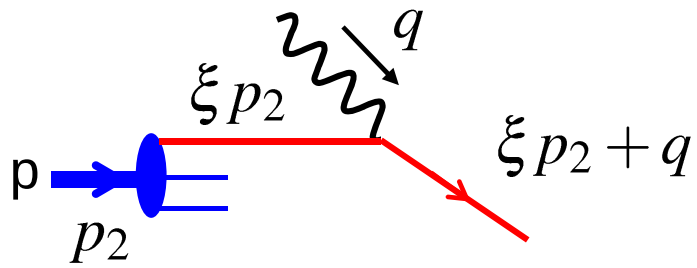
$$(\xi p_2 + q)^2 = m_q^2 \approx 0 \quad \Rightarrow \quad \cancel{\xi^2 p_2^2} + q^2 + 2\xi p_2.q = 0 \qquad (\xi^2 p_2^2 = m_q^2 \approx 0)$$

$$\Rightarrow \quad \xi = \frac{Q^2}{2p_2.q} = x$$

**Bjorken *x* can be identified as the fraction of the proton momentum carried by the struck quark (in a frame where the proton has very high energy)**

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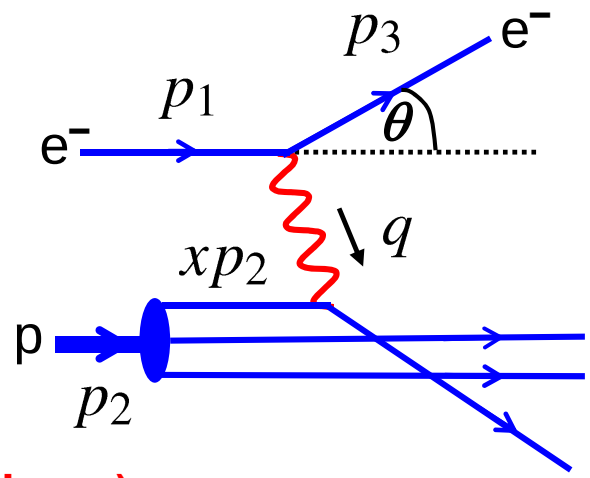
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- **But for the underlying quark interaction**

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**Now apply this to** $e^-q \to e^-q$

$$\frac{d\sigma}{dq^2} = \frac{2\pi\alpha^2 e_q^2}{q^4}\left[1 + \left(1 + \frac{q^2}{s_q}\right)^2\right]$$

**$e_q$ is quark charge, i.e.** $e_u = +2/3; \quad e_d = -1/3$

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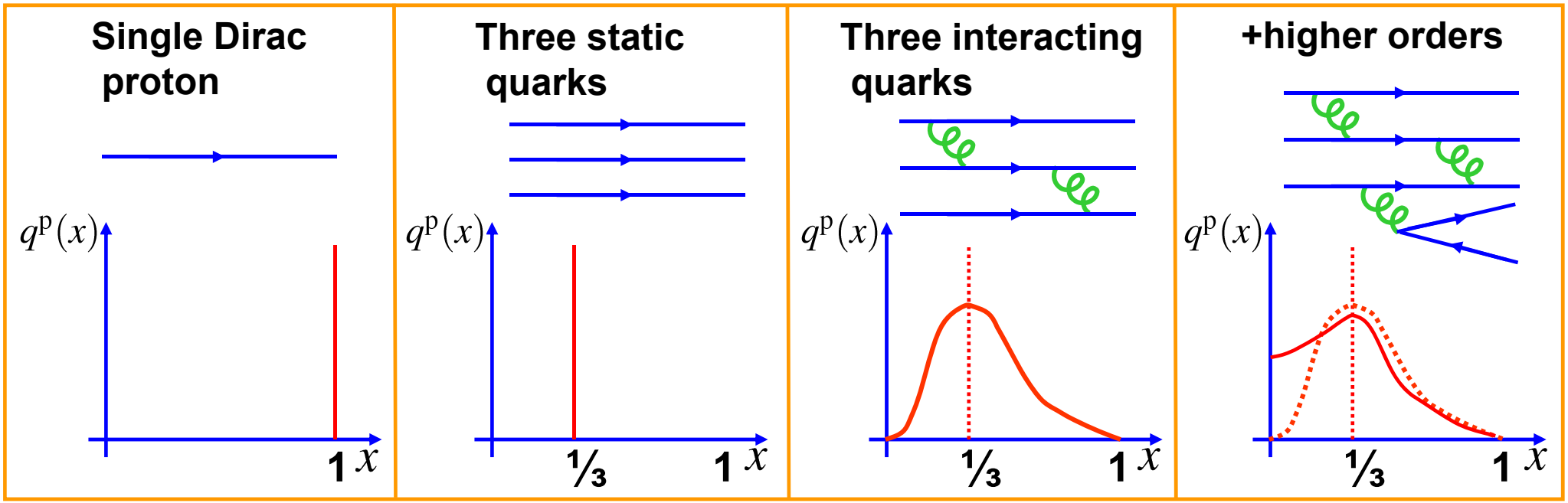
$$\frac{d\sigma}{dQ^2} = \frac{4\pi\alpha^2 e_q^2}{Q^4}\left[(1-y)+\frac{y^2}{2}\right] \qquad (3)$$

★ This is the expression for the differential cross-section for **elastic** $e^- q$ scattering from a quark carrying a fraction $x$ of the proton momentum.

• Now need to account for distribution of quark momenta within proton

★ Introduce parton distribution functions such that $q^p(x)dx$ is the number of quarks of type $q$ within a proton with momenta between $x \to x + dx$

● Expected **form of the parton distribution function ?**



★ The cross section for scattering from a **particular quark type** within the proton which in the range $x \to x + \mathrm{d}x$ is

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}Q^2} = \frac{4\pi\alpha^2}{Q^4}\left[(1-y)+\frac{y^2}{2}\right] \times e_q^2 q^{\mathrm{p}}(x)\mathrm{d}x$$

★ Summing over all types of quark within the proton gives the expression for the electron-proton scattering cross section

$$\frac{\mathrm{d}^2\sigma^{\mathrm{ep}}}{\mathrm{d}x\mathrm{d}Q^2} = \frac{4\pi\alpha^2}{Q^4}\left[(1-y)+\frac{y^2}{2}\right]\sum_q e_q^2 q^{\mathrm{p}}(x) \qquad (5)$$

★ Compare with the electron-proton scattering cross section in terms of structure functions (equation (2) ):

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}Q^2} = \frac{4\pi\alpha^2}{Q^4}\left[(1-y)\frac{F_2(x,Q^2)}{x}+y^2F_1(x,Q^2)\right] \qquad (6)$$

★ By comparing (5) and (6) obtain the parton model prediction for the structure functions in the general L.I. form for the differential cross section

$$F_2^{\mathrm{p}}(x,Q^2) = 2xF_1^{\mathrm{p}}(x,Q^2) = x\sum_q e_q^2 q^{\mathrm{p}}(x)$$

⇒ Can relate measured structure functions to the underlying quark distributions

## The parton model predicts:

- **Bjorken Scaling** $F_1(x,Q^2) \rightarrow F_1(x) \qquad F_2(x,Q^2) \rightarrow F_2(x)$
  - ✶ Due to scattering from point-like particles within the proton
- **Callan-Gross Relation** $F_2(x) = 2xF_1(x)$
  - ✶ Due to scattering from spin half Dirac particles where the magnetic moment is directly related to the charge; hence the "electro-magnetic" and "pure magnetic" terms are fixed with respect to each other.

★ At present parton distributions cannot be calculated from QCD

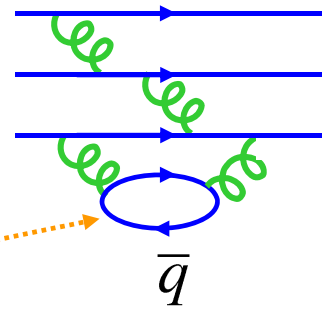
- Can't use perturbation theory due to large coupling constant

★ Measurements of the structure functions enable us to determine the parton distribution functions !

★ For electron-proton scattering we have:

$$F_2^{\mathrm{p}}(x) = x\sum_q e_q^2 q^{\mathrm{p}}(x)$$

- Due to higher orders, the proton contains not only up and down quarks but also anti-up and anti-down quarks (will neglect the small contributions from heavier quarks)



•**For electron-proton scattering have:**

$$F_2^{ep}(x) = x\sum_q e_q^2 q^p(x) = x\left(\frac{4}{9}u^p(x) + \frac{1}{9}d^p(x) + \frac{4}{9}\overline{u}^p(x) + \frac{1}{9}\overline{d}^p(x)\right)$$

•**For electron-neutron scattering have:**

$$F_2^{en}(x) = x\sum_q e_q^2 q^n(x) = x\left(\frac{4}{9}u^n(x) + \frac{1}{9}d^n(x) + \frac{4}{9}\overline{u}^n(x) + \frac{1}{9}\overline{d}^n(x)\right)$$

★**Now assume "isospin symmetry", i.e. that the neutron (ddu) is the same as a proton (uud) with up and down quarks interchanged, i.e.**

$$d^n(x) = u^p(x); \qquad u^n(x) = d^p(x)$$

**and define the neutron distributions functions in terms of those of the proton**

$$u(x) \equiv u^p(x) = d^n(x); \qquad d(x) \equiv d^p(x) = u^n(x)$$
$$\overline{u}(x) \equiv \overline{u}^p(x) = \overline{d}^n(x); \qquad \overline{d}(x) \equiv \overline{d}^p(x) = \overline{u}^n(x)$$

**giving:**

$$F_2^{ep}(x) = 2xF_1^{ep}(x) = x\left(\frac{4}{9}u(x) + \frac{1}{9}d(x) + \frac{4}{9}\overline{u}(x) + \frac{1}{9}\overline{d}(x)\right) \quad (7)$$

$$F_2^{en}(x) = 2xF_1^{en}(x) = x\left(\frac{4}{9}d(x) + \frac{1}{9}u(x) + \frac{4}{9}\overline{d}(x) + \frac{1}{9}\overline{u}(x)\right) \quad (8)$$

•**Integrating (7) and (8) :**

$$\int_0^1 F_2^{ep}(x)\mathrm{d}x = \int_0^1 x\left(\frac{4}{9}[u(x)+\bar{u}(x)]+\frac{1}{9}[d(x)+\bar{d}(x)]\right)\mathrm{d}x = \frac{4}{9}f_u+\frac{1}{9}f_d$$

$$\int_0^1 F_2^{en}(x)\mathrm{d}x = \int_0^1 x\left(\frac{4}{9}[d(x)+\bar{d}(x)]+\frac{1}{9}[u(x)+\bar{u}(x)]\right)\mathrm{d}x = \frac{4}{9}f_d+\frac{1}{9}f_u$$

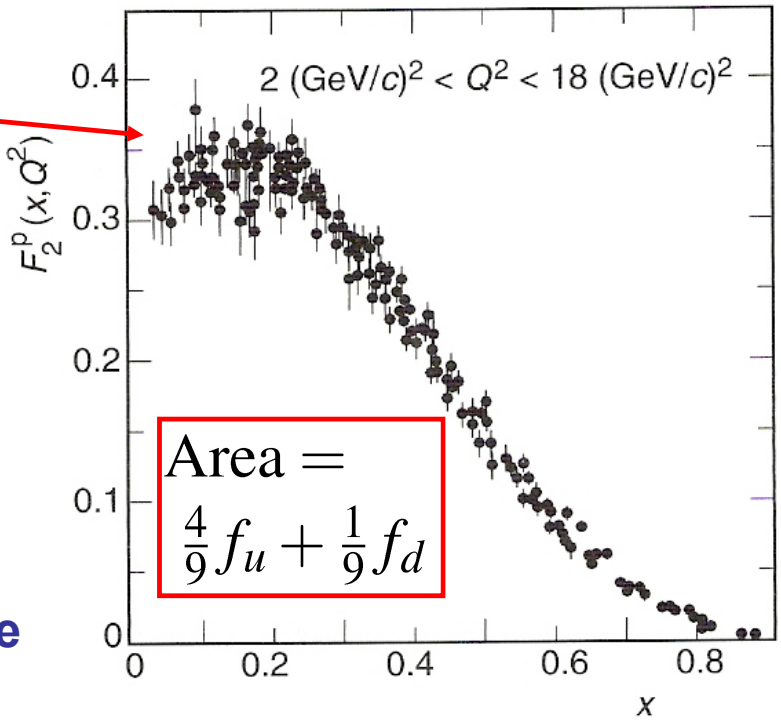
★ $f_u = \int_0^1 [xu(x)+x\bar{u}(x)]\mathrm{d}x$ **is the fraction of the proton momentum carried by the up and anti-up quarks**

**Experimentally**

$$\int F_2^{ep}(x)\mathrm{d}x \approx 0.18$$

$$\int F_2^{en}(x)\mathrm{d}x \approx 0.12$$

➡ $f_u \approx 0.36 \quad f_d \approx 0.18$

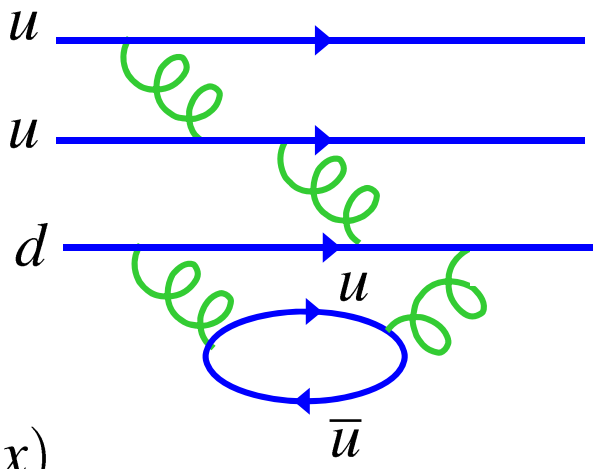



✶ **In the proton, as expected, the up quarks carry twice the momentum of the down quarks**

✶ **The quarks carry just over 50 % of the total proton momentum. The rest is carried by gluons (which being neutral doesn't contribute to electron-nucleon scattering).**

# Valence and Sea Quarks

- As we are beginning to see the proton is complex…
- The parton distribution function $u^{\mathrm{p}}(x) = u(x)$ includes contributions from the "valence" quarks and the virtual quarks produced by gluons: the "sea"




- Resolving into valence and sea contributions:

$$u(x) = u_{\mathrm{V}}(x) + u_{\mathrm{S}}(x) \qquad d(x) = d_{\mathrm{V}}(x) + d_{\mathrm{S}}(x)$$
$$\bar{u}(x) = \bar{u}_{\mathrm{S}}(x) \qquad \bar{d}(x) = \bar{d}_{\mathrm{S}}(x)$$

- The proton contains two valence up quarks and one valence down quark and would expect:

$$\int_0^1 u_{\mathrm{V}}(x)\mathrm{d}x = 2 \qquad \int_0^1 d_{\mathrm{V}}(x)\mathrm{d}x = 1$$

- But no *a priori* expectation for the total number of sea quarks !
- But sea quarks arise from gluon quark/anti-quark pair production and with $m_u = m_d$ it is reasonable to expect

$$u_{\mathrm{S}}(x) = d_{\mathrm{S}}(x) = \bar{u}_{\mathrm{S}}(x) = \bar{d}_{\mathrm{S}}(x) = S(x)$$

- With these relations (7) and (8) become

$$F_2^{\mathrm{ep}}(x) = x\left(\frac{4}{9}u_{\mathrm{V}}(x) + \frac{1}{9}d_{\mathrm{V}}(x) + \frac{10}{9}S(x)\right) \qquad F_2^{\mathrm{en}}(x) = x\left(\frac{4}{9}d_{\mathrm{V}}(x) + \frac{1}{9}u_{\mathrm{V}}(x) + \frac{10}{9}S(x)\right)$$

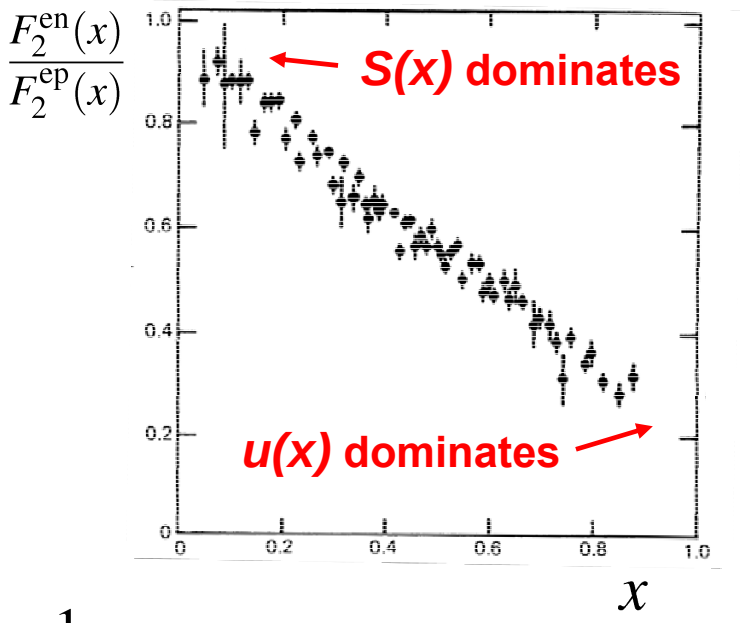
Giving the ratio

$$\frac{F_2^{en}(x)}{F_2^{ep}(x)} = \frac{4d_V(x) + u_V(x) + 10S(x)}{4u_V(x) + d_V(x) + 10S(x)}$$

- The sea component arises from processes such as $g \to \bar{u}u$ . Due to the $1/q^2$ dependence of the gluon propagator, much more likely to produce low energy gluons. Expect the sea to comprise of low energy $q/\bar{q}$
- Therefore at low *X* expect the sea to dominate:

$$\frac{F_2^{en}(x)}{F_2^{ep}(x)} \to 1 \quad \text{as} \quad x \to 0$$

**Observed experimentally**




- At high *X* expect the sea contribution to be small

$$\frac{F_2^{en}(x)}{F_2^{ep}(x)} \to \frac{4d_V(x) + u_V(x)}{4u_V(x) + d_V(x)} \quad \text{as} \quad x \to 1$$

**Note:** $u_V = 2d_V$ would give ratio 2/3 as $x \to 1$

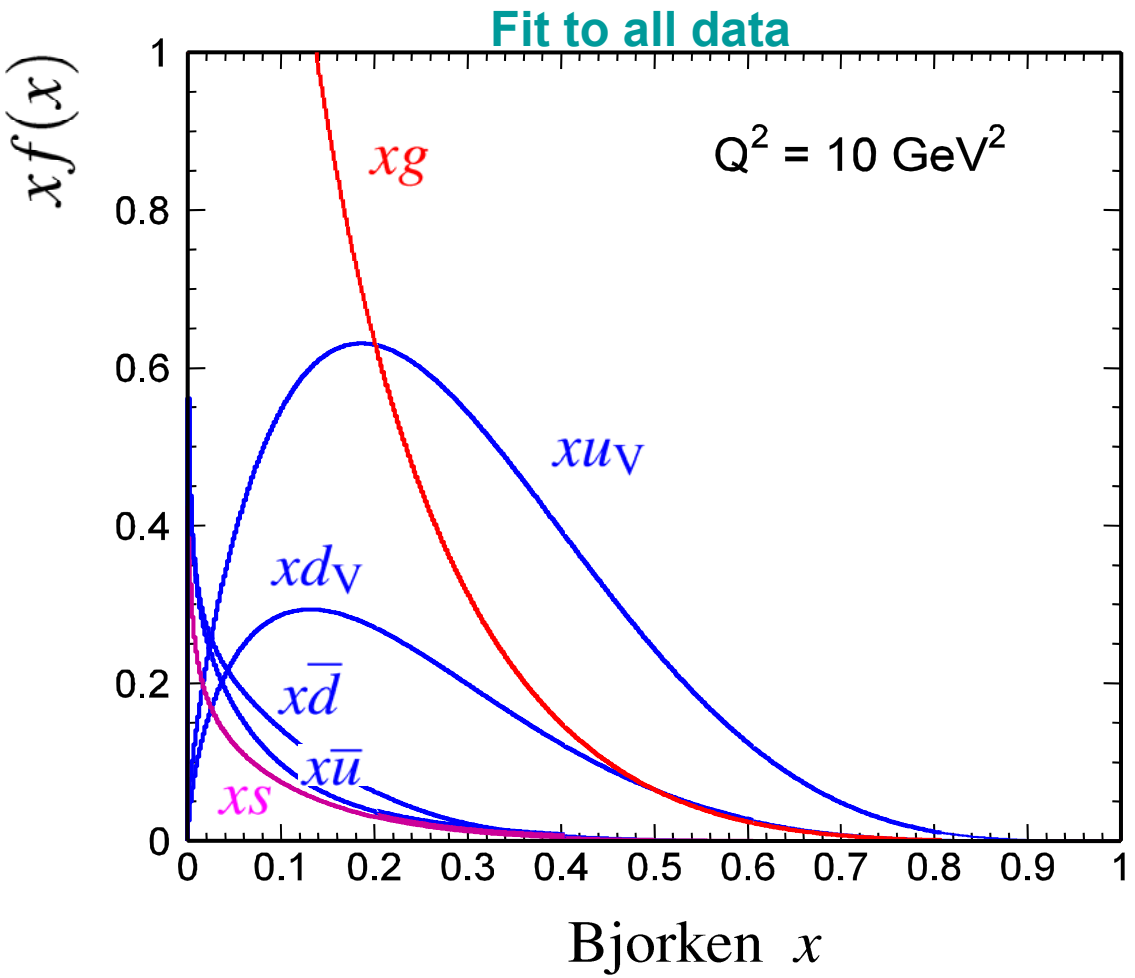
**Experimentally** $F_2^{en}(x)/F_2^{ep}(x) \to 1/4 \quad \text{as} \quad x \to 1$

$$\Rightarrow \quad d(x)/u(x) \to 0 \quad \text{as} \quad x \to 1$$

**This behaviour is not understood.**

# Parton Distribution Functions

- **Ultimately the parton distribution functions are obtained from a fit to all experimental data including neutrino scattering (see handout 10)**
  - **Hadron-hadron collisions give information on gluon pdf** $g(x)$




**Note:**

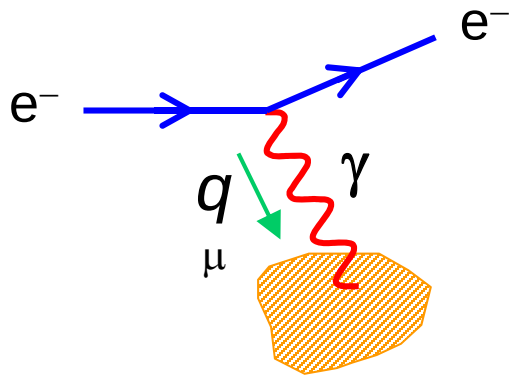
- **Apart from at large** $x$
  $$u_V(x) \approx 2d_V(x)$$
- **For** $x < 0.2$
  **gluons dominate**
- **In fits to data assume**
  $$u_s(x) = \bar{u}(x)$$
- $\bar{d}(x) > \bar{u}(x)$
  **not understood – exclusion principle?**
- **Small strange quark component** $s(x)$

**(Try Question 12 )**

# Scaling Violations

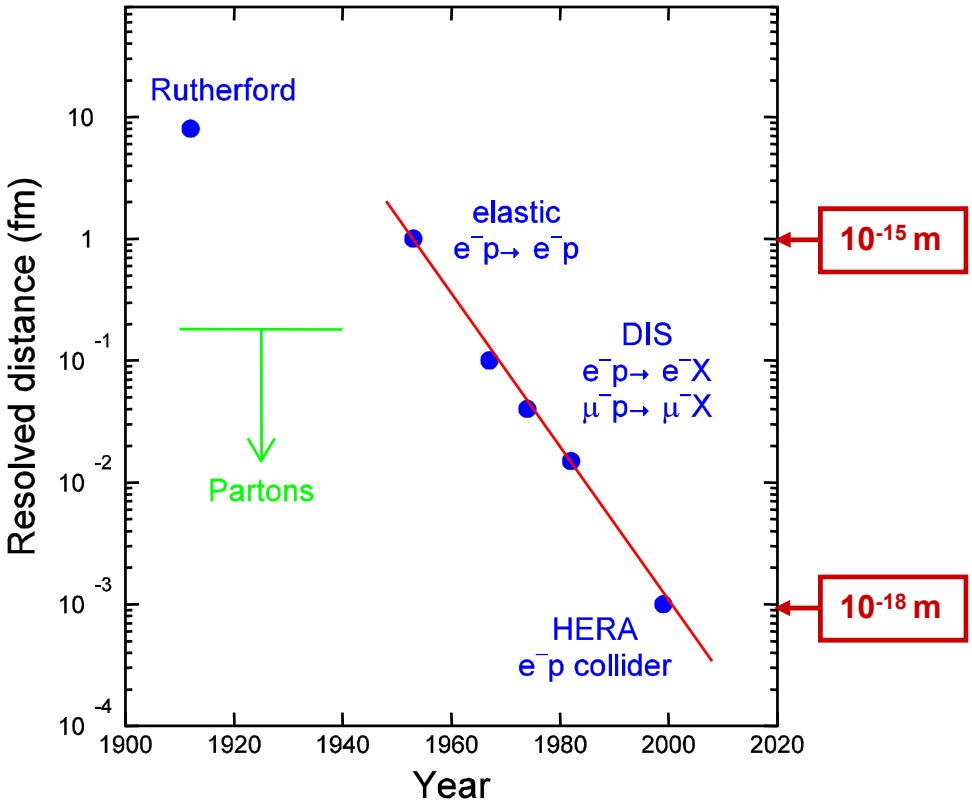
- In last 40 years, experiments have probed the proton with virtual photons of ever increasing energy
- Non-point like nature of the scattering becomes apparent when $\lambda_\gamma$ ~ size of scattering centre

$$\lambda_\gamma = \frac{h}{|\vec{q}|} \sim \frac{1\,\text{GeV fm}}{|\vec{q}|(\text{GeV})}$$




- Scattering from point-like quarks gives rise to **Bjorken scaling**: no $q^2$ cross section dependence
- IF quarks were not point-like, at high $q^2$ (when the wavelength of the virtual photon ~ size of quark) would observe rapid decrease in cross section with increasing $q^2$.
- To search for quark sub-structure want to go to highest $q^2$

**HERA**



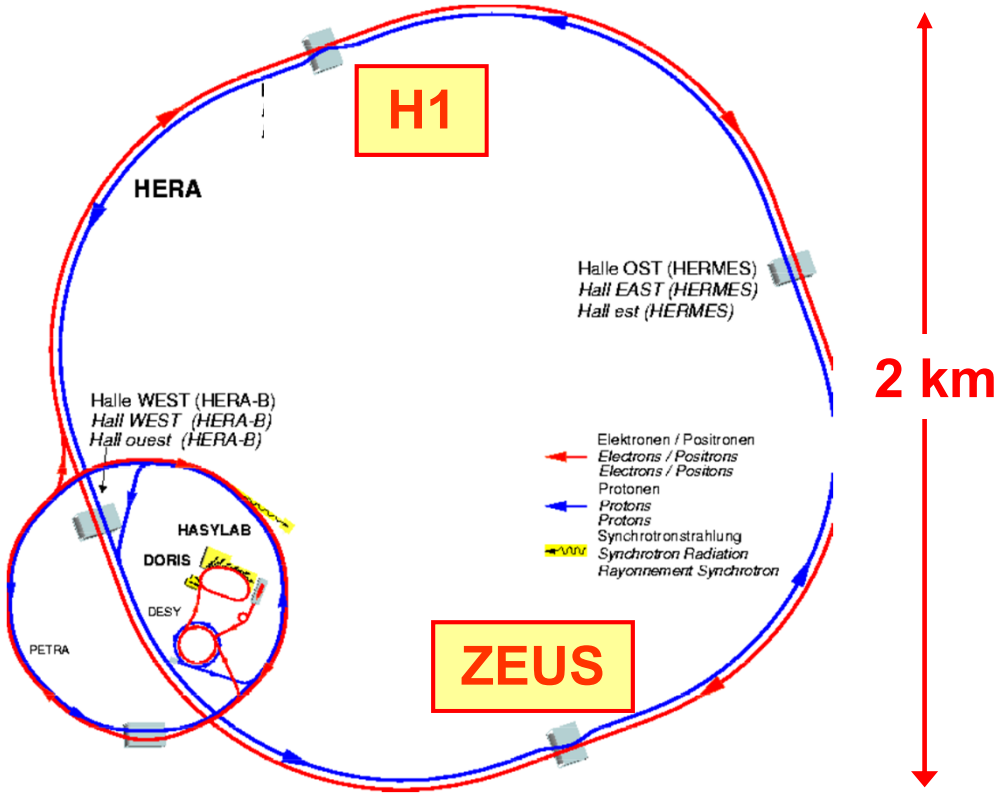
# HERA $e^{\pm}p$ Collider : 1991-2007

★ DESY (Deutsches Elektronen-Synchroton) Laboratory, Hamburg, Germany

$e^{\pm}$ **27.5 GeV** → ← **820 GeV** p

**$\sqrt{s}$ = 300 GeV**



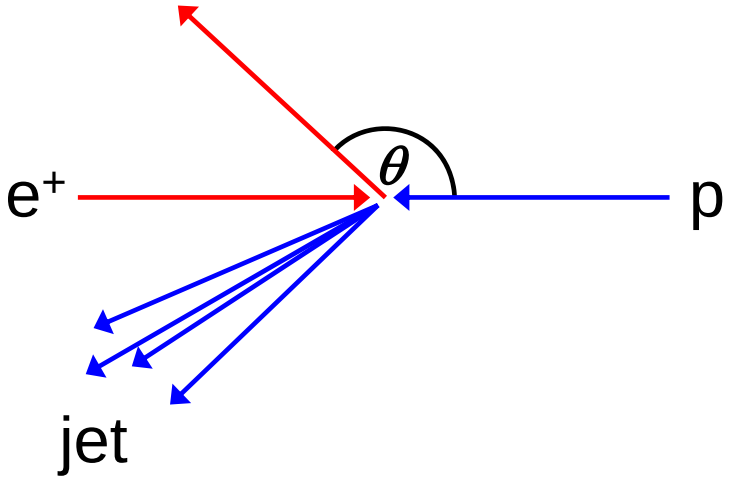




★ Two large experiments : H1 and ZEUS

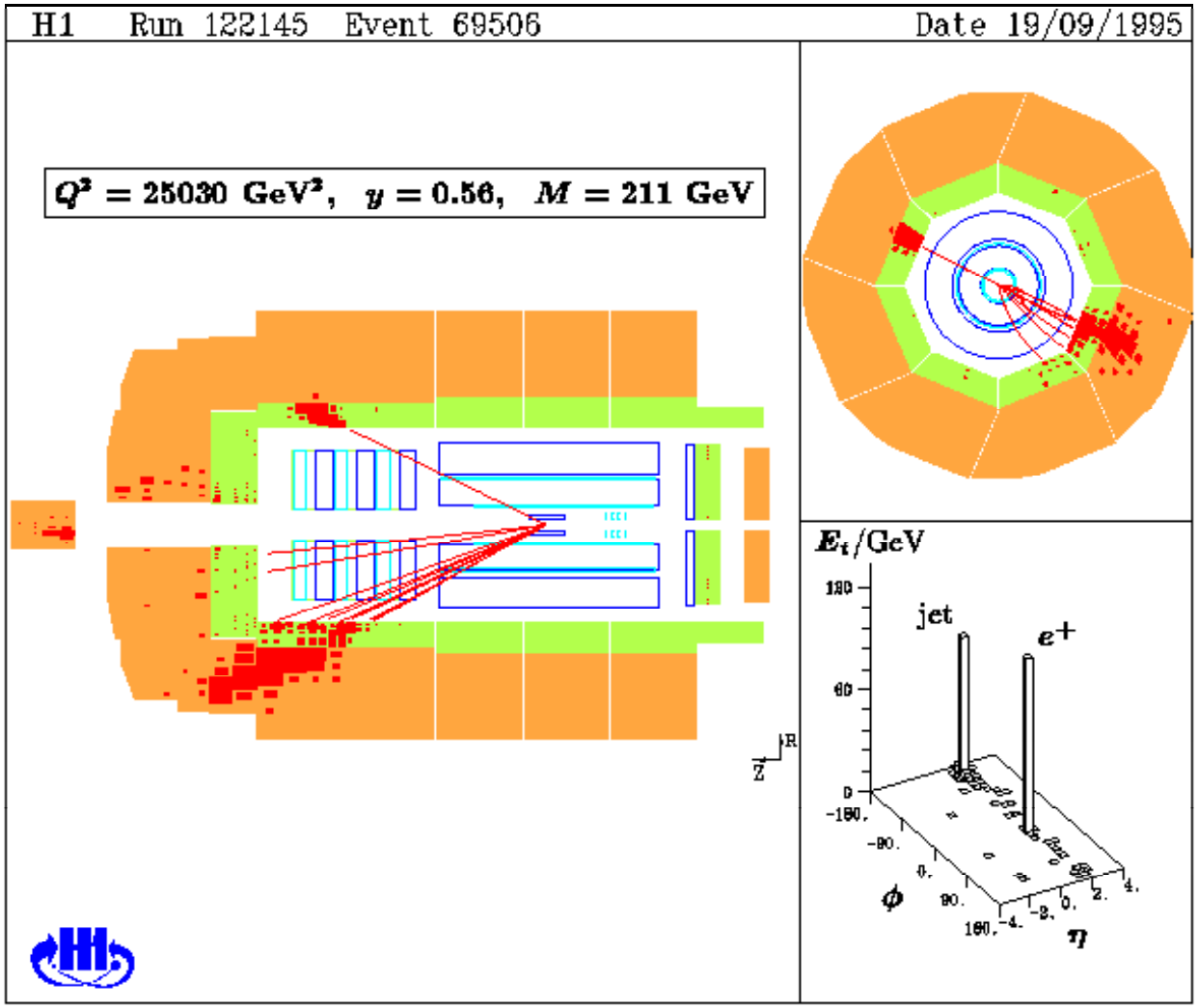
★ Probe proton at very high $Q^2$ and very low $x$

# Example of a High $Q^2$ Event in H1

- Event kinematics determined from electron angle and energy




- Also measure hadronic system (although not as precisely) – gives some redundancy



# $F_2(x,Q^2)$ Results

★ **No evidence of rapid decrease of cross section at highest $Q^2$**

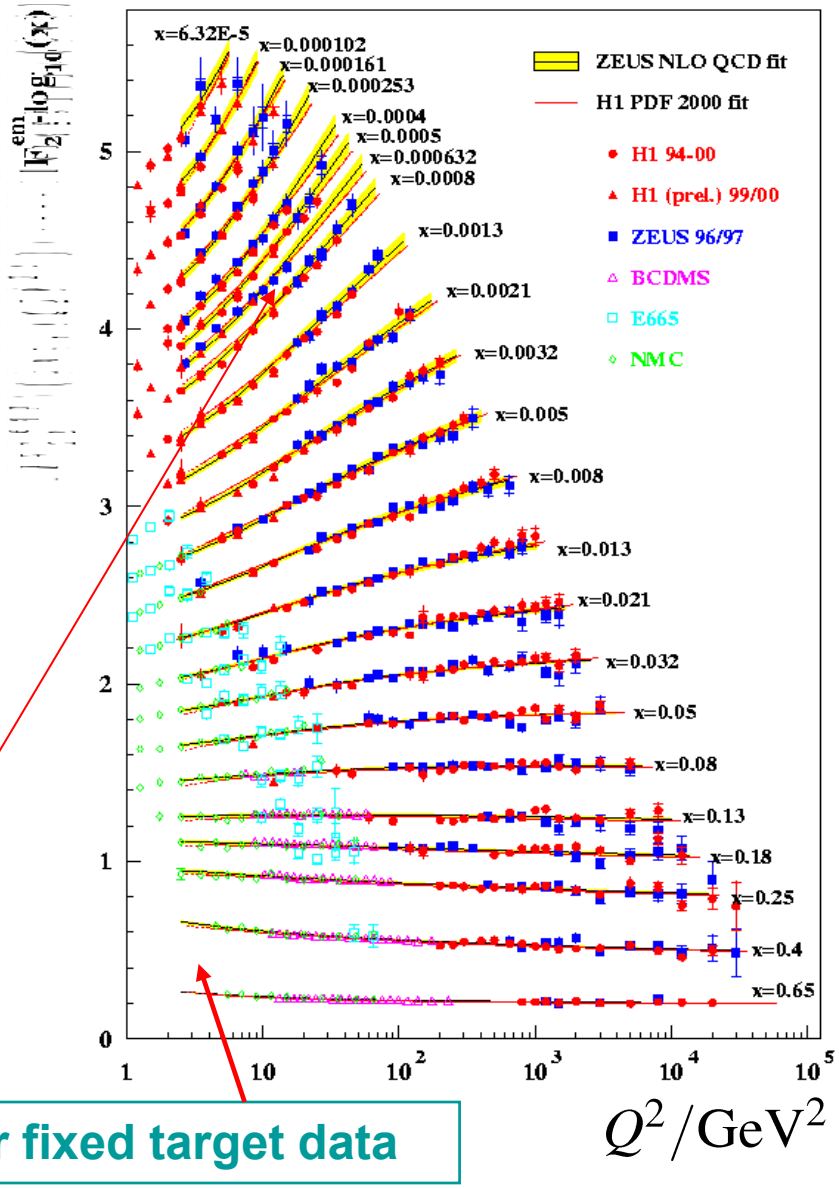
➡ $R_{\text{quark}} < 10^{-18}\,\text{m}$

★ **For $x > 0.05$, only weak dependence of $F_2$ on $Q^2$ : consistent with the expectation from the quark-parton model**

★ **But observe clear scaling violations, particularly at low $x$**

$$F_2(x,Q^2) \neq F_2(x)$$

**Interesting "negative" size at very small x?**




**Earlier fixed target data**

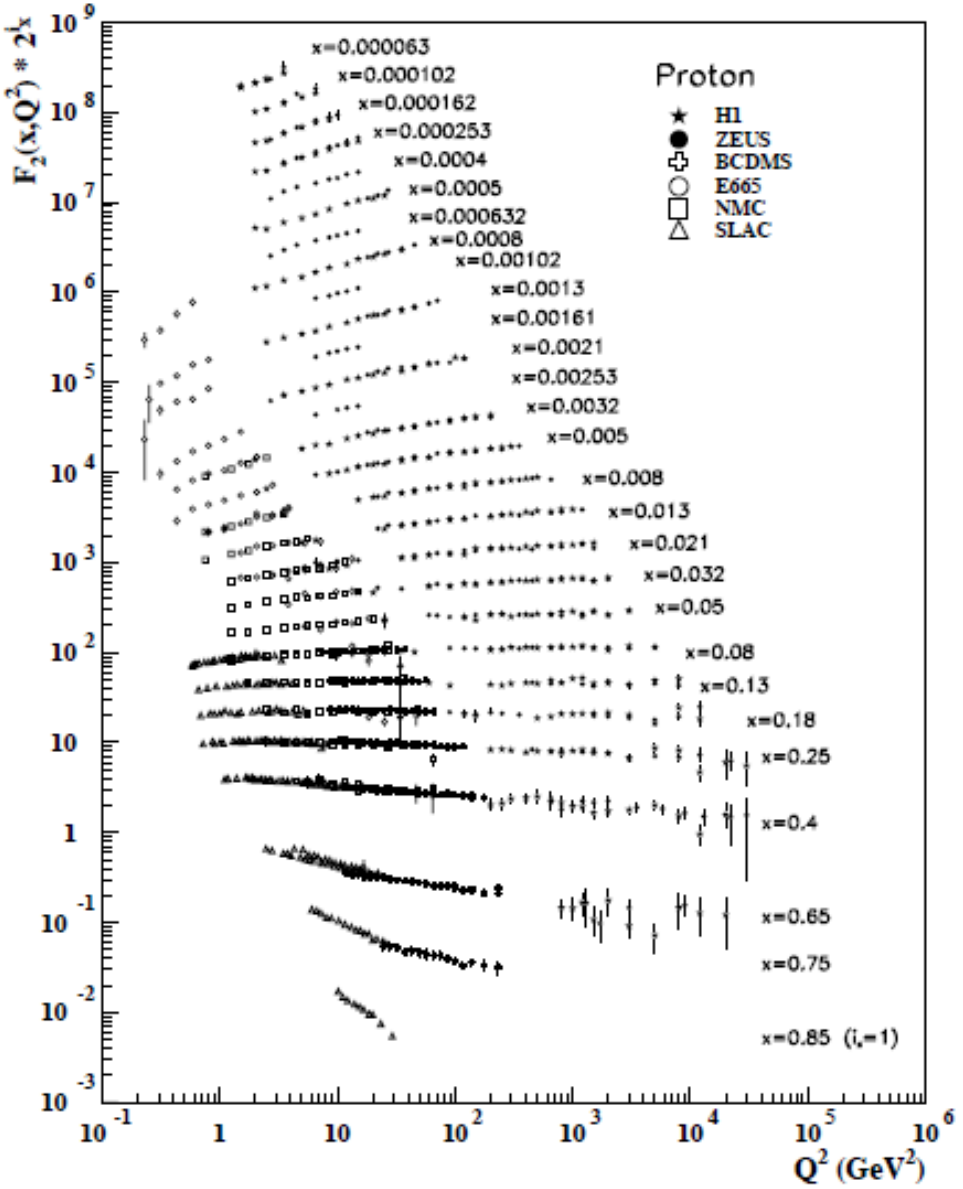
# Level to which they are $Q^2$ independent

- World data

On structure fn's

$$F_2 \propto \sum_i q_i f_i(x)$$

Not exaxtly q^2 independent ?
"Scaling Violations"



# FLAT BECAUSE Elastic POINTLIKE

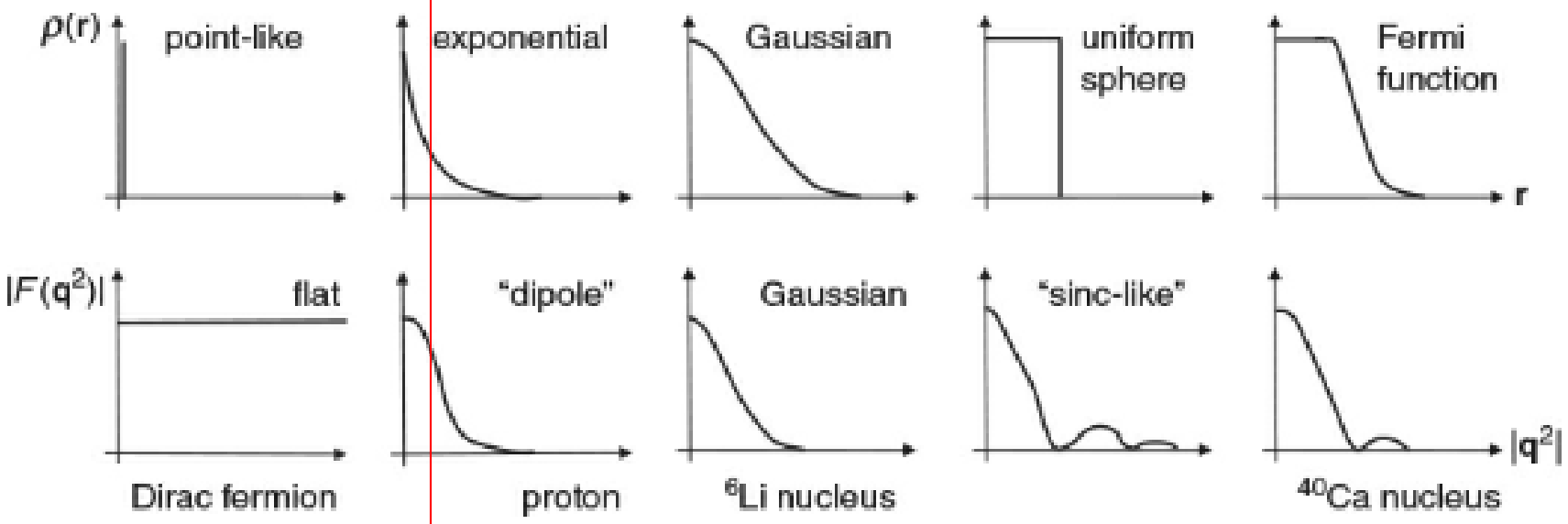
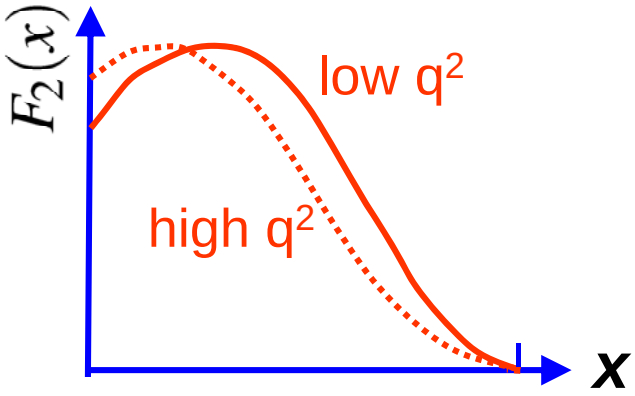
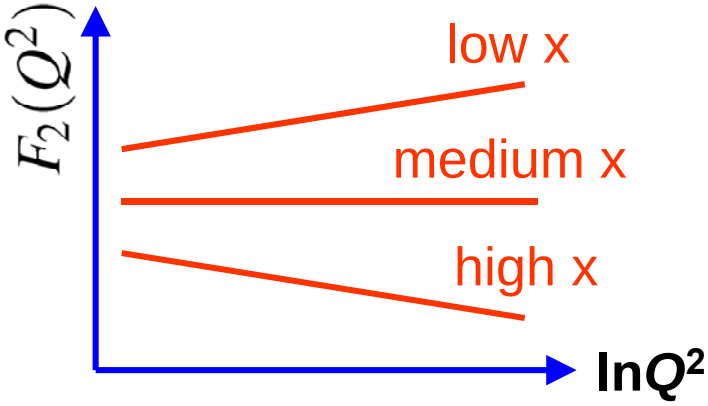



**Fig. 7.5** Possible three-dimensional charge distributions and the corresponding form factors plotted as a function of $\mathbf{q}^2$.

# Origin of Scaling Violations

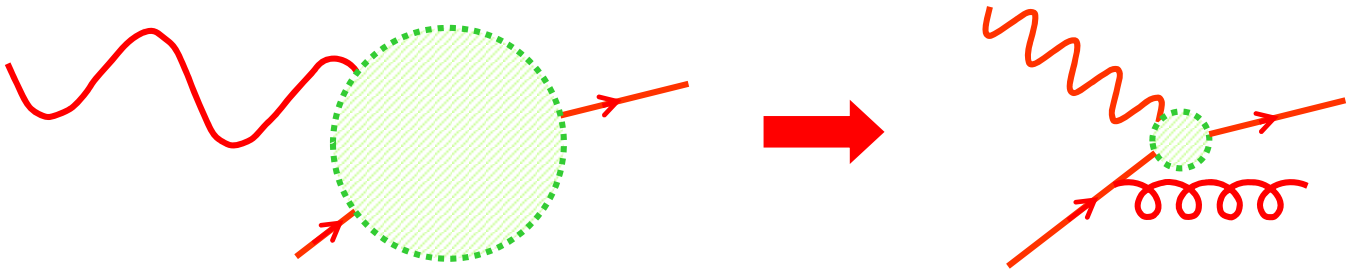
★ Observe "small" deviations from exact Bjorken scaling $F_2(x) \rightarrow F_2(x, Q^2)$




★ At high $Q^2$ observe more low $x$ quarks

★ "Explanation": at high $Q^2$ (shorter wave-length) resolve finer structure: i.e. reveal quark is sharing momentum with gluons. At higher $Q^2$ expect to "see" more low $x$ quarks



★ QCD cannot predict the $x$ dependence of $F_2(x, Q^2)$

★ But QCD can predict the $Q^2$ dependence of $F_2(x, Q^2)$

# DGLAP/BFKL Struct Fn Evolution

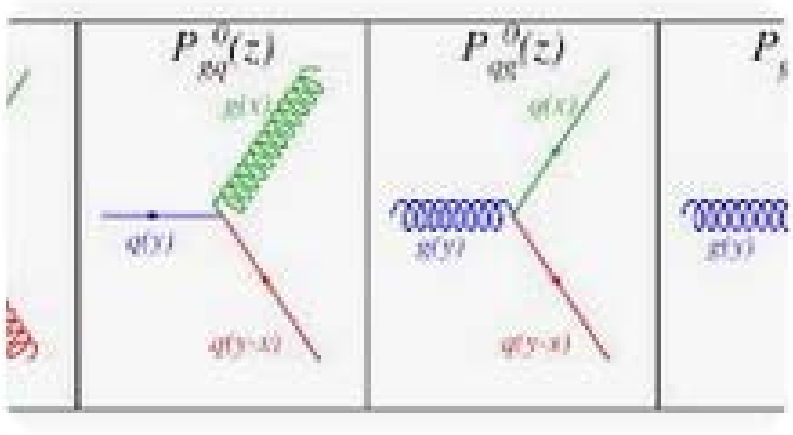
- DGLAP pQCD "Evolution" takes f(x,q1^2) to another q2
- Uses q/g "splitting functions" Can be predicted from **Pert.**QCD

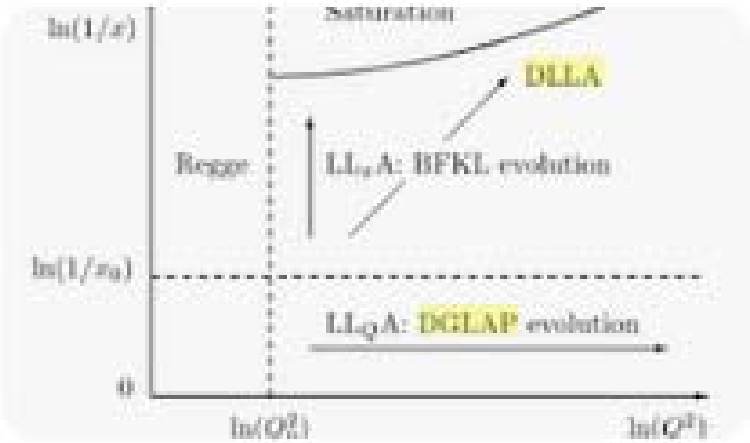
$$\frac{dg(x,Q^2)}{d\ln Q^2} = \frac{\alpha_s(Q^2)}{2\pi}\int_0^1 \frac{dy}{y}\left[\Sigma_q P_{gq}(z) q(y,Q^2) + P_{gg}(z) g(y,Q^2)\right]$$

$$P_{gq} \to \frac{C_F}{z}, \qquad P_{gg} \to \frac{2C_A}{z}$$

$$\frac{dg(x,Q^2)}{d\ln Q^2} = \frac{\alpha_s(Q^2)}{2\pi}\int_x^1 \frac{dy}{y}\frac{6}{z} g(y,Q^2)$$

$$\begin{aligned}
P_{q_i q_j}(z,\alpha_s) &= \delta_{ij} P_{qq}^{(0)}(z) + \frac{\alpha_s}{2\pi} P_{q_i q_j}^{(1)}(z) + \ldots \\
P_{qg}(z,\alpha_s) &= P_{qg}^{(0)}(z) + \frac{\alpha_s}{2\pi} P_{qg}^{(1)}(z) + \ldots \\
P_{gq}(z,\alpha_s) &= P_{gq}^{(0)}(z) + \frac{\alpha_s}{2\pi} P_{gq}^{(1)}(z) + \ldots \\
P_{gg}(z,\alpha_s) &= P_{gg}^{(0)}(z) + \frac{\alpha_s}{2\pi} P_{gg}^{(1)}(z) + \ldots
\end{aligned}$$




- BFKL evolution is similar thing but x1→x2 evolution

# From devations from Q^2 const: g(x)

- World data

On structure fn's

Not exaxtly q^2 independent ?
"Scaling Violations"

Can derive actual g(x) structure fn from "GLOBAL" fit parameters describing scaling violations

